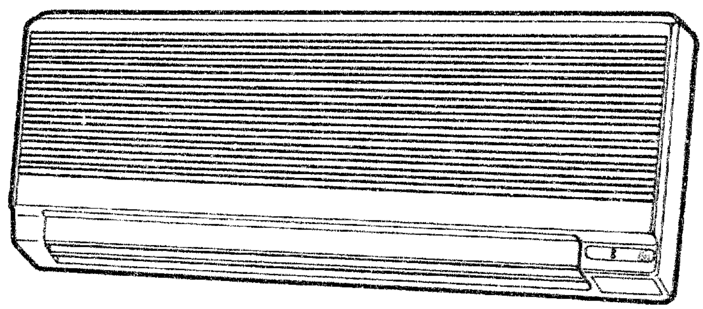
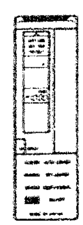
SERVICE DATA
FILE NO. 300-787

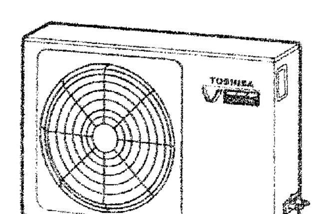
TOSHIBA

ROOM AIR-CONDITIONER
[SPLIT TYPE]

RAS-45BKHV(W)/BAHV







CONTENTS

1. SPECIFICATIONS

MODEL NAME: RAS – 45BKHV(W)/BAHV				
Items		Unit	Cooling	Heating
Capacity *1		kcal/h	4250	6000
		Btu/h	17000	24000
Power source		Phase	Single	
		V	220/240	
		Hz	50	
Power consumption		kW	2.45	2.87
Power factor		%	90	90
Running current		A	12.40/11.40	14.50/13.30
Starting current		A	7	7
Moisture removal		lit/h	2.6	
Noise	Indoor unit (H/M/L)	dB	46/42/38	
	Outdoor unit	dB	52	
Refrigerant	Name of refrigerant		R–22	
	Charge volume	kg (oz)	1.15 (40.6)	
Refrigerant control			Capillary tube & Expansion valve	
Inter-connection pipe	Gas side size	mm (in)	12.7(1/2)	
	Connection type		Flare connection	
	Liquid side size	mm (in)	6.35 (1/4)	
	Connection type		Flare connection	
	Maximum length (of one way) *2	m (ft)	15 (49.2)	
	Maximum height indoor unit higher	m (ft)	6 (19.7)	
	Maximum height Outdoor unit higher	m (ft)	6 (19.7)	
Condensate drain pipe	Outer diameter	mm (in)	16 (5/8)	

Specifications are subject to change without notice.

INDOOR UNIT		Units	RAS-45BKHV(W)	
Dimensions	Height	mm (in.)	370 (14.6)	
	Width	mm (in.)	960 (37.8)	
	Depth	mm (in.)	17.5 (6.9)	
Net weight		kg (lbs)	13 (29.25)	
Evaporator type			Finned tube	
Indoor fan type			Cross flow fan	
Air volume	High fan	m^3/h (CFM)	732 (431)	
	Medium fan	m^3/h (CFM)	———	
	Low fan	m^3/h (CFM)	———	
Fan motor output		W	30	
Air filter			Polyproprene Net filter (Washable)	
OUTDOOR UNIT		Units	45BAHV	
Dimensions	Height	mm (in.)	675 (26.6)	
	Width	mm (in.)	720 (28.3)	
	Depth	mm (in.)	300 + 23 (11.8 + 0.9)	
Net weight		kg (lbs)	55 (121.3)	
Condenser type			Finned tube	
Outdoor fan type			Propeller	
Fan motor output		W	40	
Com-pressor	Model		HV340X2-S12L1	
	Output	W	1100	
Safety device			Fuse, compressor thermal relay	
Auto louver			Yes	
Outdoor temperature range for proper performance			21 ~ 43 (70 ~ 109)	−5 ~ 21 (23 ~ 70)
Usable outdoor temperature range		°C (°F)	10 ~ 43 (50 ~ 109)	−10 ~ 21 (14 ~ 70)

Specifications are subject to change without notice.

Note1:

Capacity is based on the following temperature condition.

		JIS	
		COOLING	HEATING
Indoor unit inlet air temperature	(DB)	27 °C (80 °F)	21 °C (70 °F)
	(WB)	19.5 °C (67 °F)	———
Outdoor unit inlet air temperature	(DB)	35 °C (95 °F)	7 °C (45 °F)
	(WB)	24 °C (75 °F)	6 °C (43 °F)

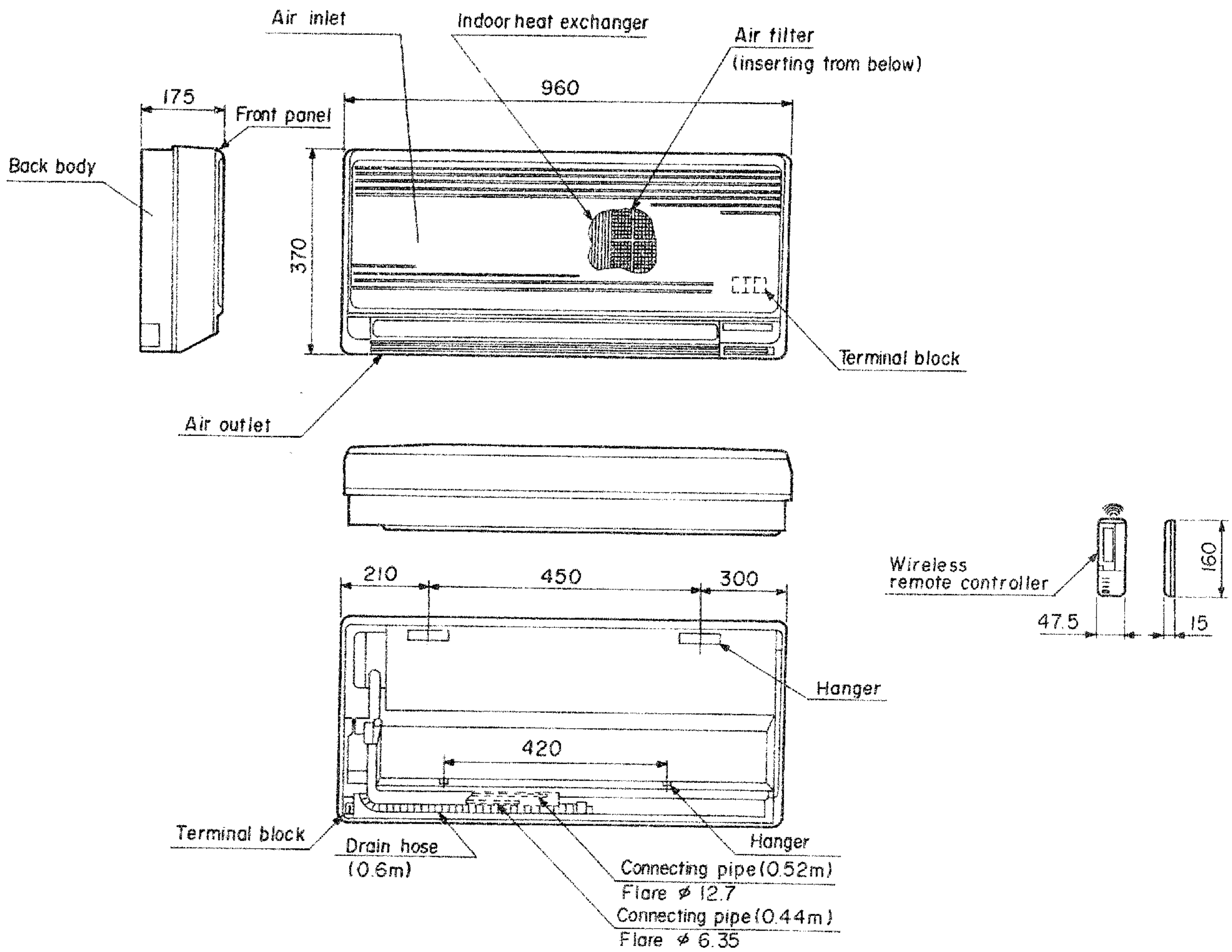
Note2:

These mean equivalent length.

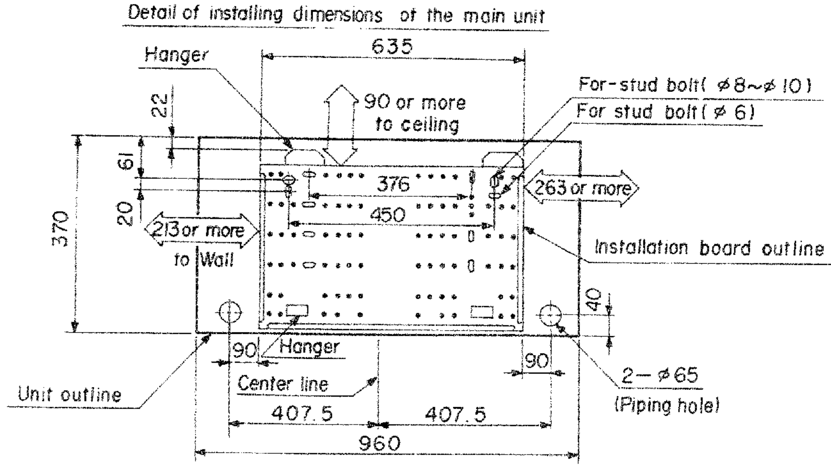
When piping length exceeds 5 m, it is necessary to add refrigerant. Additional 10 g needs per extention of 1 m.

2. CONSTRUCTION VIEWS

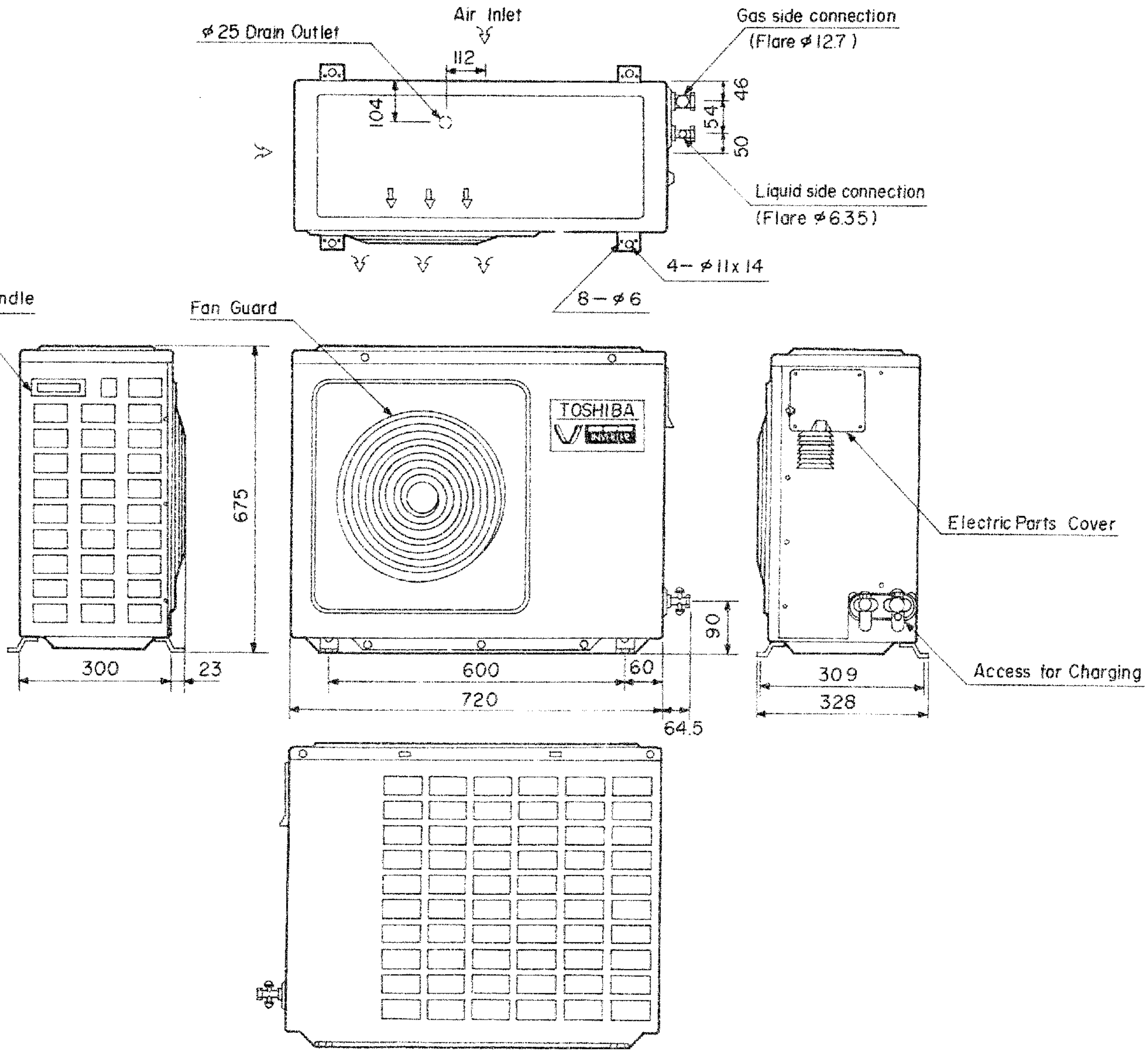
Indoor unit (RAS-45BKHV(W))

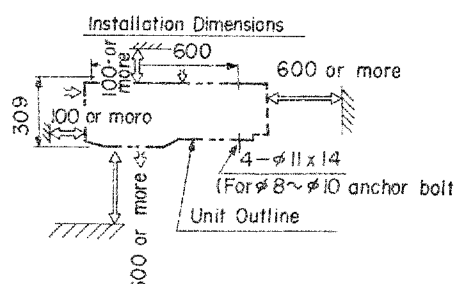


Detail of installing dimensions of the main unit

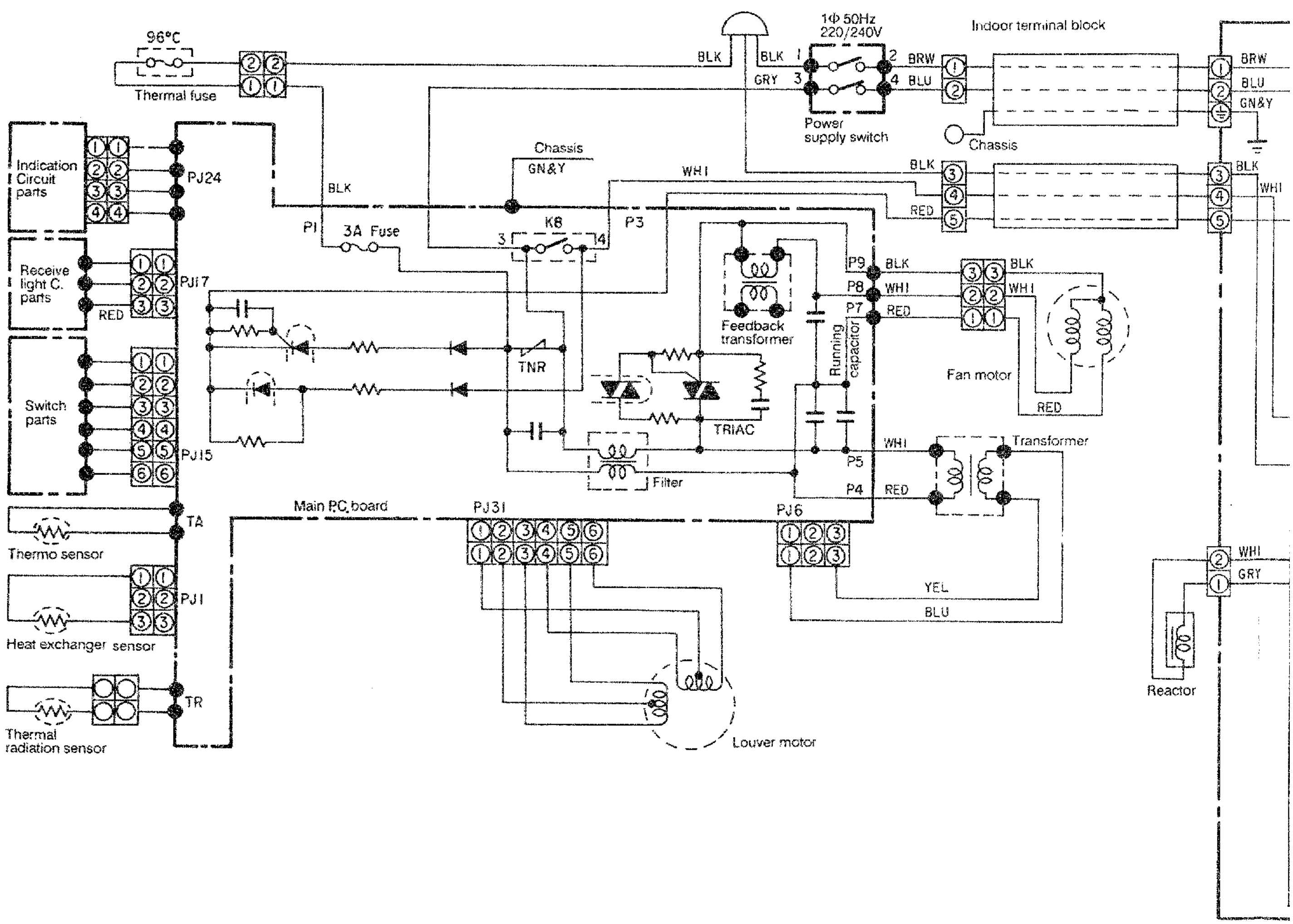


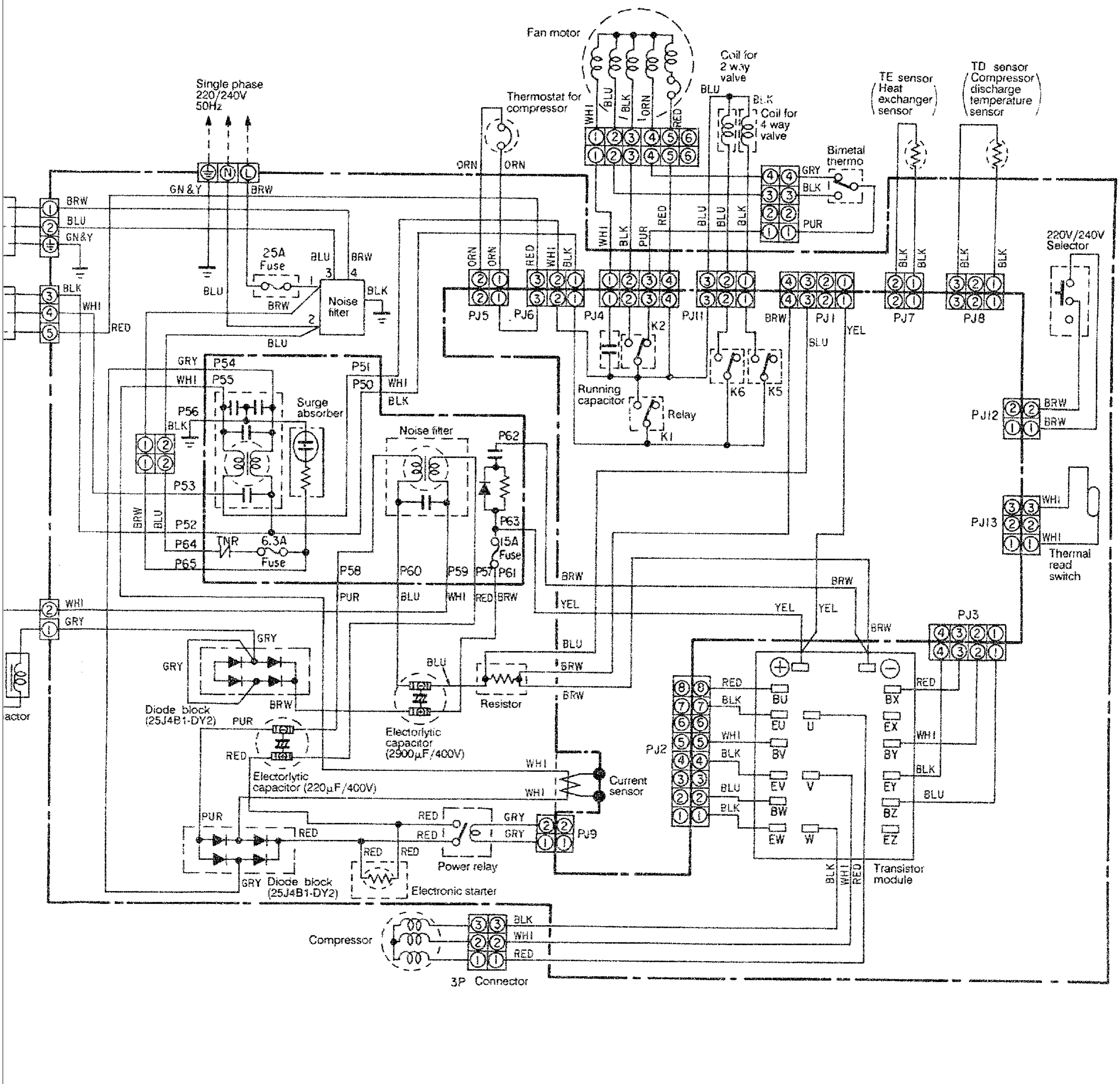
Outdoor unit (RAS-45BAHV)





3. WIRING DIAGRAM (RAS-45BKHV(W)/BAHV)





Fan motor
Single phase 220/240V 50Hz
Thermostat for compressor
Coil for 2 way valve
Coil for 4 way valve
TE sensor Heat exchanger sensor
TD sensor Compressor discharge temperature sensor
Bimetal thermo
25A Fuse
Noise filter
220V/240V Selector
Surge absorber
Running capacitor
Relay
Noise filter
15A Fuse
6.3A Fuse
Thermal read switch
Diode block (25J4B1-DY2)
Electorlytic capacitor (2900μF/400V)
Resistor
Electorlytic capacitor (220μF/400V)
Current sensor
Power relay
Electronic starter
Transistor module
Compressor
3P Connector

4. SPECIFICAITONS OF ELECTRICAL PARTS

RAS-45BKHV(W) (Indoor unit)

No.	Parts name	Type	Specifications		
1	Fan motor (for indoor)	AF-230-30-4D	Output (Rated) 30W, 4 pole, 1 phase, 220/240V, 50Hz		
			Winding resistance (Ω) at 20 °C)	Red-black	White-black
				221	200
2	Transformer	FT-25	Primary : 220/240V, Secondary : 13V		
3	Microprocessor	MCC-1080	TMP47C860N-2852Z		
4	Power relay (K8)	G4F1112TP	Coil: DC 12V, 75mA Contact: AC 250V, 35/10A		
5	TRIAC	SM8JZ46	8A, 600V		
6	Line filter	UF-103Y0R6	10mH, AC 0.6A		
7	Ceramic capacitor for AC	DE7150-1FZ103P	AC 400V, 0.01μF		
8	Heat exchanger sensor	(Microprocessor)	10kΩ (at 25°C)		
9	Thermo. (room temperature) sensor	(Microprocessor)	10kΩ (at 25°C)		
10	Remote controller	(Remote control)	————		
11	Running capacitor	(EEP2G185HQN 105)	1.8μF, 400V		
12	Terminal block	————	3P, AC 300V, 20A		
13	Terminal block	————	2P, AC 300V, 20A		
14	Power supply switch	————	AC 250V, 20A		
15	Louver motor	————	Output (Rated) 2W, 10 pole, 1 phase, DC 12V, 50Hz		
			Winding resistance (Ω) (at 20 °C)	Blue-brown, Red-pink, Yellow-brown, Red-orange	
				160	
16	Feedback transformer	PTM230 – 10S – 04	Primary : 240V, Secondary : 12V		

Specifications are subject to change without notice.

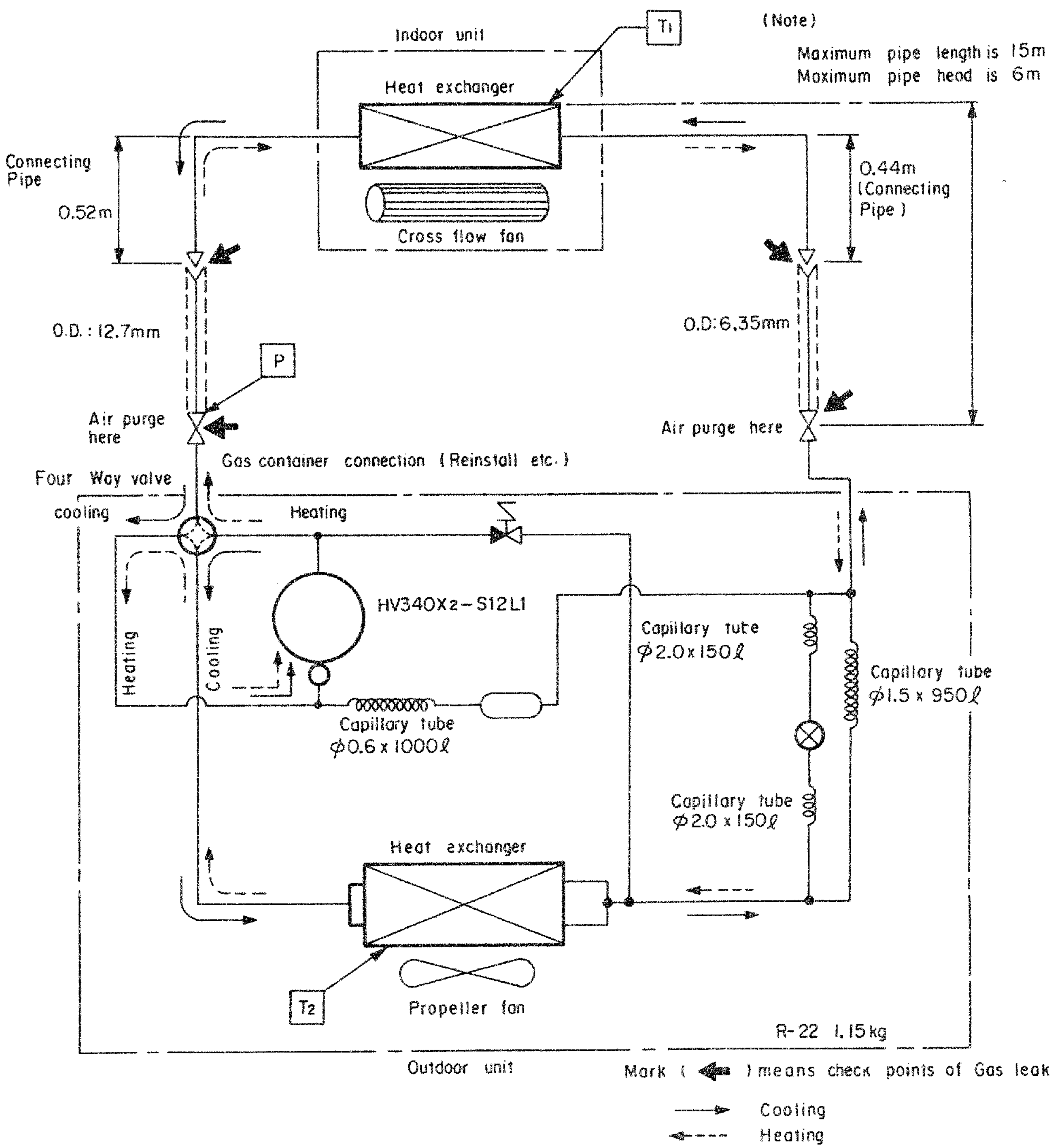
RAS-45BAHV (Outdoor unit)

<table>
<tr><th>No.</th><th>Parts name</th><th>Type</th><th colspan="3">Specification</th></tr>
<tr><td rowspan="3">1</td><td rowspan="3">Compressor</td><td rowspan="3">HV340X2-S12L1</td><td colspan="3">Output (Rated)750W, 2 poles, 3 phase, 114V, 60Hz</td></tr>
<tr><td rowspan="2">Winding resistance (Ω) (at 20°C)</td><td>Red-Black</td><td>White-Black</td></tr>
<tr><td>0.59</td><td>0.59</td></tr>
<tr><td rowspan="3">2</td><td rowspan="3">Fan motor (For outdoor)</td><td rowspan="3">AF-230-50L</td><td colspan="3">Output (Rated) 50W, 6 poles, 1 phase, 230V, 50Hz</td></tr>
<tr><td rowspan="2">winding resistance (Ω) (at 20°C)</td><td>Red-Black</td><td>White-Black</td></tr>
<tr><td>116</td><td>128</td></tr>
<tr><td>3</td><td>Running capacitor</td><td>EEP2W305HWN1</td><td colspan="3">3.0μF, AC 450V</td></tr>
<tr><td>4</td><td>Compressor thermal relay (Thermostat)</td><td>CS-7</td><td colspan="3">OFF: 125 ±4°C, ON: 90 ± 5°C</td></tr>
<tr><td>5</td><td>4-way valve</td><td>V26-110B</td><td colspan="3">Coil: AC 220/240V</td></tr>
<tr><td>6</td><td>2-way valve</td><td>NEV603DXF</td><td colspan="3">Coil: AC 220/240V</td></tr>
<tr><td>7</td><td>Relay for fan motor (K1, K2, K5)</td><td>G4C-112P</td><td colspan="3">Coil: DC 12V Contact: AC 240V, 1.7A</td></tr>
<tr><td>8</td><td>Relay for 4-way valve and 2-way valve (K5, K6)</td><td>G4U-112P</td><td colspan="3">Coil: DC 12V Contact: AC 240V, 0.8A</td></tr>
<tr><td>9</td><td>Bimetal thermostat</td><td>UT10-164</td><td colspan="3"></td></tr>
<tr><td>10</td><td>SC coil (Line filter)</td><td>NF20KL151 or TF3233S-151Y20R0</td><td colspan="3">20A</td></tr>
<tr><td>11</td><td>SC coil (Line filter)</td><td>TF6029S-102Y20R0</td><td colspan="3">20A</td></tr>
<tr><td>12</td><td>DC-DC transformer</td><td>SWT-10A</td><td colspan="3">Primary: DC 350V, Secondary: 0-7-13V, 0-5.5V x 4</td></tr>
<tr><td>13</td><td>Reactor</td><td>CH-17B-4</td><td colspan="3">L=2.8mH, 18A</td></tr>
<tr><td>14</td><td>PF Terminal block</td><td>PF-3</td><td colspan="3">20A, AC 300V, 3P</td></tr>
<tr><td>15</td><td>Fuse</td><td>(for inverter overcurrent)</td><td colspan="3">25A, AC 250V</td></tr>
<tr><td>16</td><td>Electrolytic capacitor</td><td>LIN2G292 MACATF</td><td colspan="3">2900μF, 400WV</td></tr>
<tr><td>17</td><td>Electrolytic capacitor</td><td>CE62W-2G221K</td><td colspan="3">220μF, 450WV</td></tr>
<tr><td>18</td><td>Diode block</td><td>25J4B41-DY2</td><td colspan="3">25A, 600V</td></tr>
<tr><td>19</td><td>Transistor module</td><td>MG30G6EL2</td><td colspan="3">30A, 450V</td></tr>
<tr><td>20</td><td>Noise filter</td><td>ZAC2217-03</td><td colspan="3">18A, 250V</td></tr>
</table>

Specifications are subject to change without notice.

5. REFRIGERATING CYCLE DIAGRAM

RAS-45BKHV(W)/BAHV

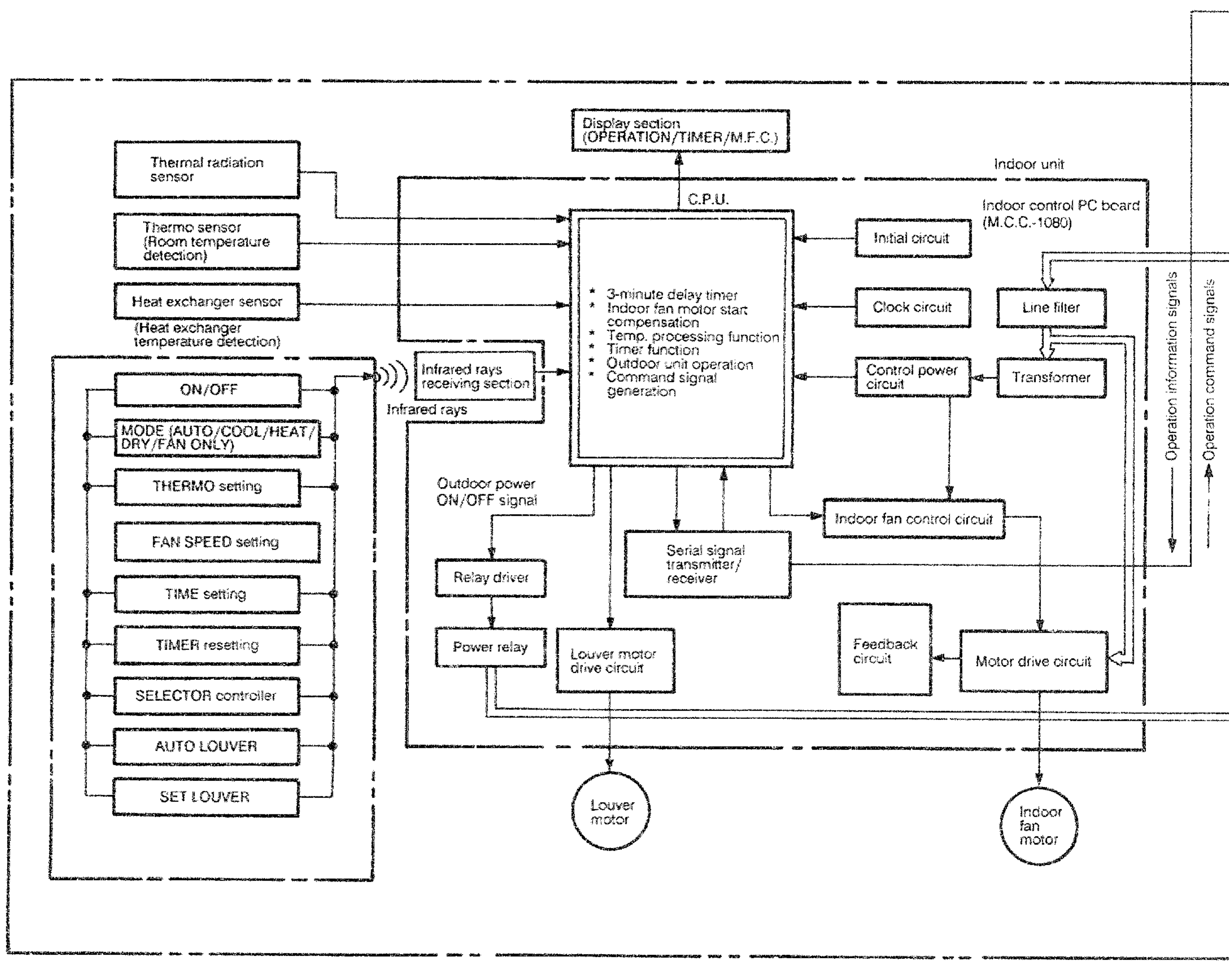


Conditions		Standard pressure P (KG.CM2G)	Surface temp. of heat exchanger interchanging pipe (°C)		Ambient temp. conditions DB.WB (°C)	
			T1	T2	Indoor	Outdoor
Cooling (80Hz)	Standard	5.3	10	46	27/19.5	35/24
	High temperature	6.2	13.5	55	32/22.5	43/25.5
	Low temperature	4.0	2.0	35	21/15.5	21/15.5
Heating (90Hz)	Standard	21.5	55	0	21/—	7/6
	High temperature	23.5	60	8	24/—	21/15.5
	Low temperature	19.2	50	– 10	21/—	– 4/– 4.5

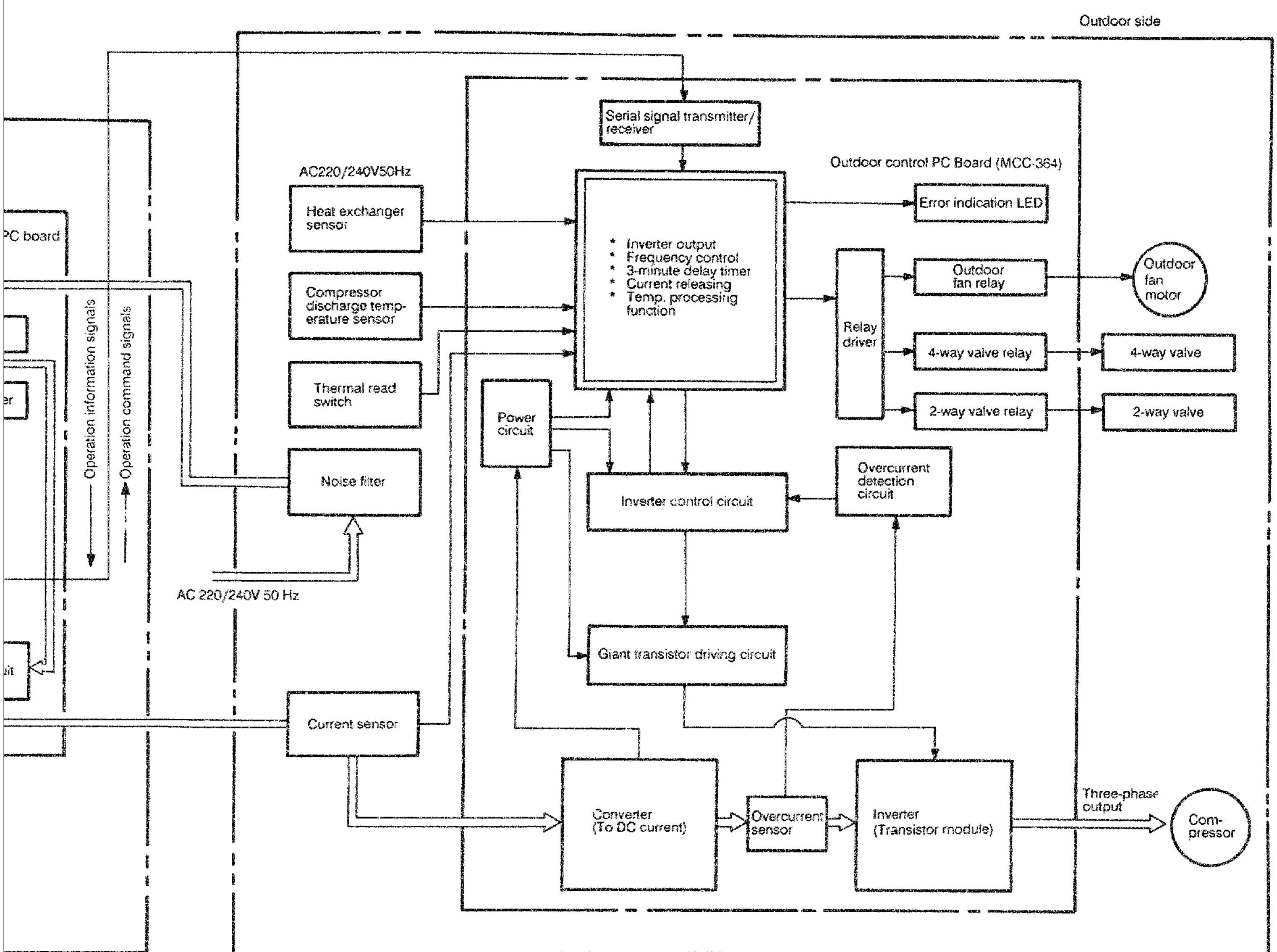
Note:
Measure the heat exchanger temperature at the center of U-bend.
(By means of thermistor thermometer)

6. MICROPROCESSOR BLOCK DIAGRAM

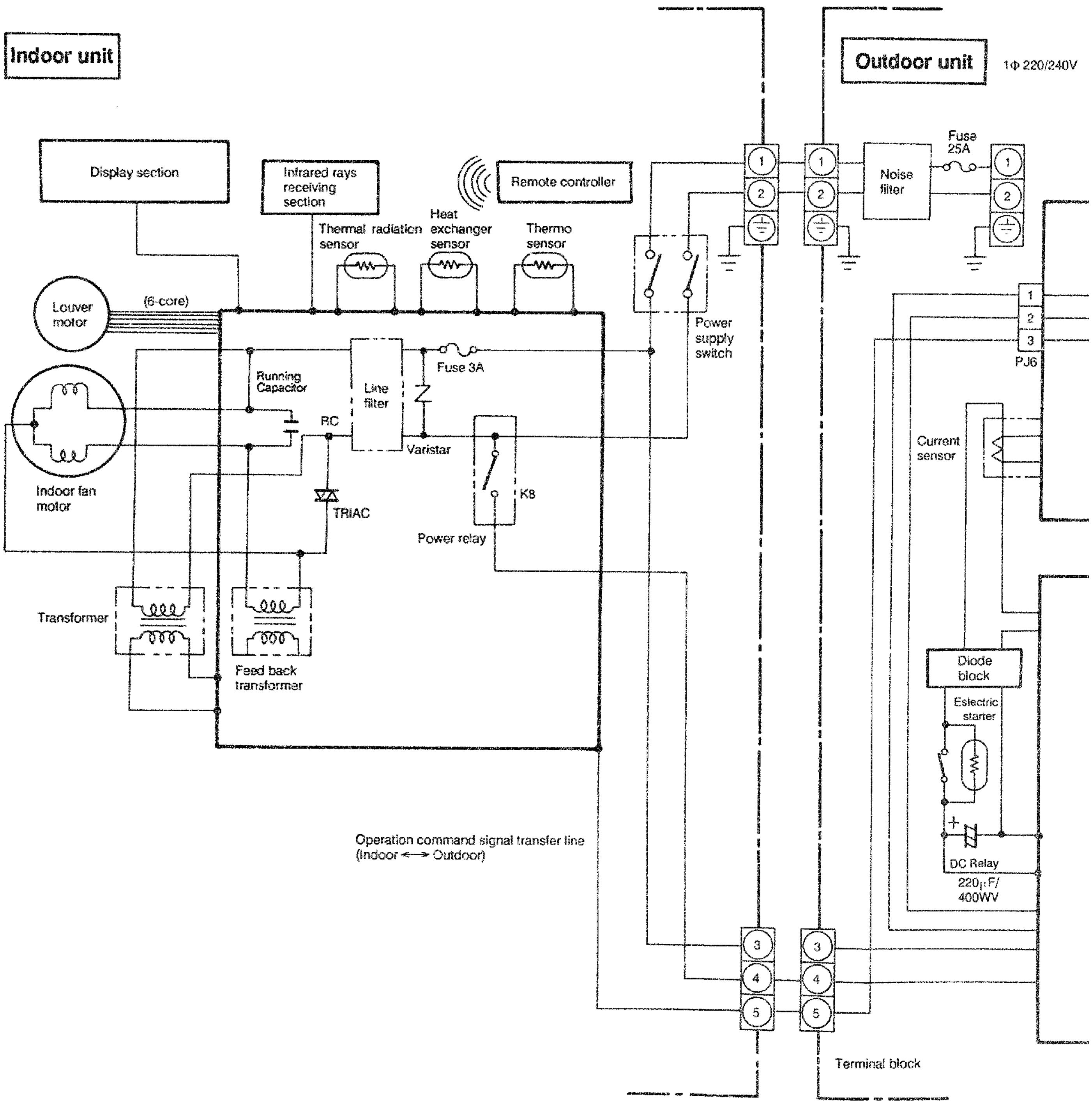
Indoor unit RAS-45BKHV (W)

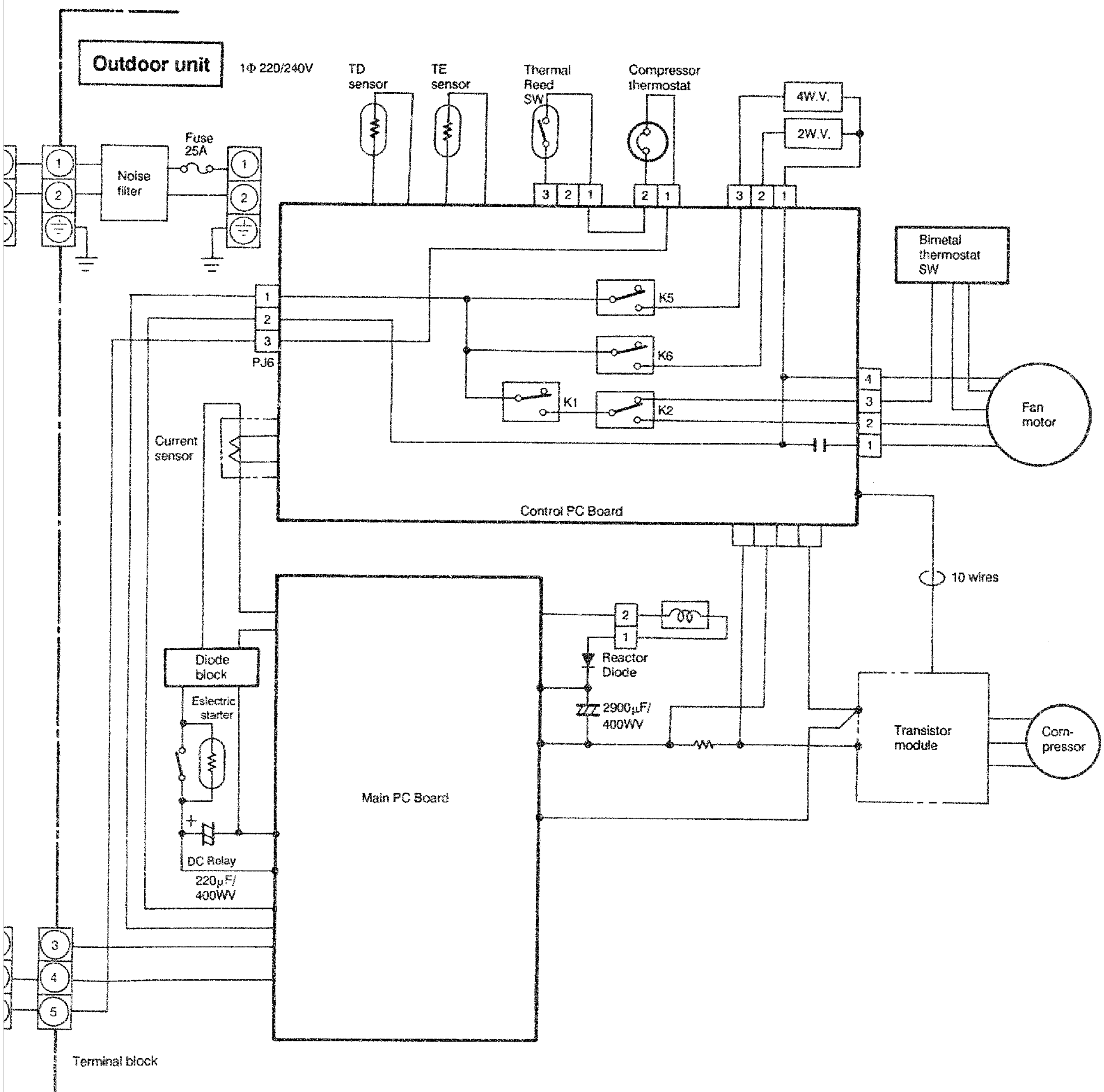


Outdoor unit RAS-45BAHV



7. CIRCUIT DIAGRAM





Outdoor unit
1Φ 220/240V
TD sensor
TE sensor
Thermal Reed SW
Compressor thermostat
4W.V.
2W.V.
Fuse 25A
Noise filter
Bimetal thermostat SW
K5
K6
K1
K2
PJ6
Current sensor
Fan motor
Control PC Board
10 wires
Reactor
Diode
2900μF/ 400WV
Diode block
Eslectric starter
Transistor module
Com- pressor
Main PC Board
DC Relay
220μF/ 400WV
Terminal block

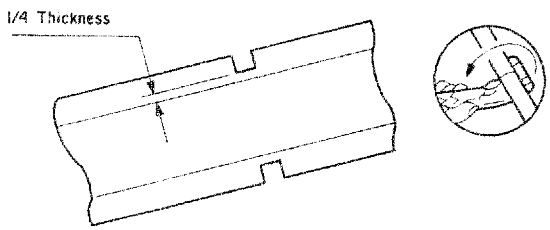
8. PIPING WORK

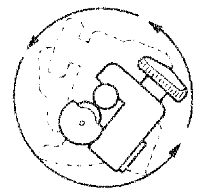
FLARING

Pipe Cutting

- Cut the pipe little by little, advancing the blade of pipe cutter slowly.

- For fine work, do not advance the blade by force in order to prevent distortion of the pipe, and extra burrs.
- Cut the pipe a little longer than required.

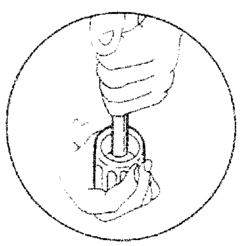


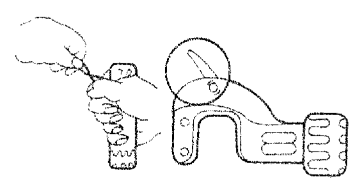


Burr Removal

- Raise burrs with the burr stick.
  Or remove burrs with the burr remover.
  (Hold the flaring end down to prevent burrs from dropping inside pipe.)

- Be sure to remove burrs to avoid unevenness on the flare face which will cause gas leaks.
- Cover the pipe end with a polyethylene bag and on sheet.

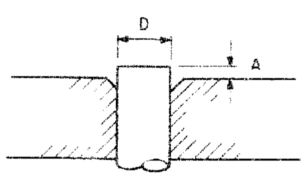




Point of Flaring

- The exact length of pipe protruding from the face of the flare die is determined by the flaring tool.

- Too long protrusion of pipe from the flaring tool will result in too long a flare; if this happens, the flare nut will not tighten and the joint will not seat., Too small protrusion will gas leaks.
  (The table below shows the use of an imperial die and a rigid die.)



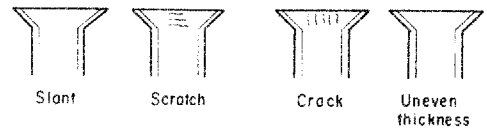
Type of pipe	D (mm)	A (mm)	
		For imperial die	For rigid die
1/4"	6.35	1.3	0.7
1/2"	12.7	1.8	1.0 ~ 1.1

CAUTION:
1. Too large a protrusion of pipe results in impossible tightening of the flare nut.
2. Too small a protrusion of pipe will cause gas leaks due to insufficient flare surface.

- Fine flaring work shows even brightness on the flared surface and uniform thickness of pipe.
- Unsuccessful flaring work causes gas leaks therefore rework it.

Finishing Work

Fix the pipe firmly on the flare die-match the centers of both the flare die and the flaring punch, and tighten flaring punch fully.

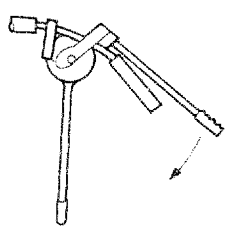


- The above faulty flaring will cause gas leaks.

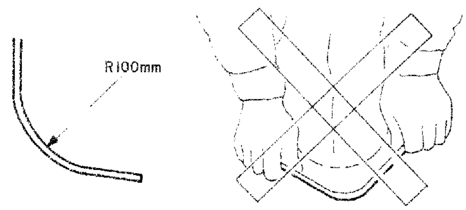
PIPE PROCESS POINTS

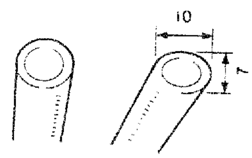
Pipe bending

- Carefully bend by hand.
- Bend with pipe bender.
  (Since the insulation is wound on, use the bender by sliding the insulation to one side, or cutting it halfway.)
- The radius of the pipe bend should be 4 inches or more.
- Too small a radius causes collapse of pipe.



→ • Collapsed pipe cause cooling failure.



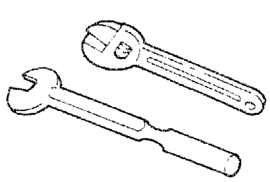


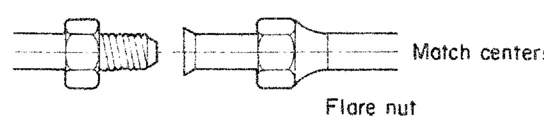
The distortion ratio must be less than 10:7.

Pipe connection

- Connect the flared pipe close to the air conditoner unit so both faces fit well.

→ • Forcing to tighten the nut without the center axis coincided causes gas leaks.





- Use a torque wrench and spanner to tighten or untighter the flare nut.
  Tightening should be done by finger first, then use the torque wrench.
  A torque wrench helps to provide adequate torque for tightening.

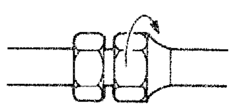
→ • Insufficient tightening causes gas leaks. Over tightening results in cracks in the flared surface.

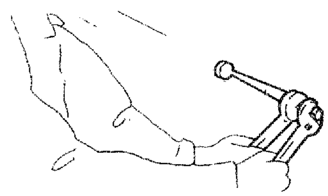
CAUTION:

1. Do not use the spanner from the beginning of tightening.
2. When processing 1/4 inch pipe, tighten lightly with spanner and then tighten additionally by 90 – 120 degrees with torque wrench and spanner.

Tightening torque of flare nut **(Unit:kgF-cm)**

Pipe size	6.35mm (1/4")	12.7mm (1/2")
Standard torque	160	500
Additional tightening	200	550

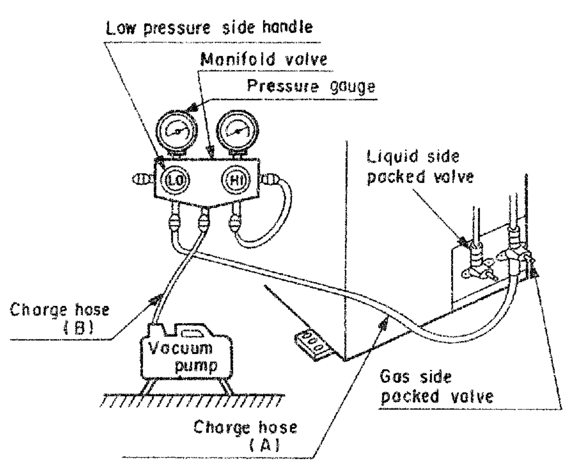




AIR PURGE

Use of vacuum pump

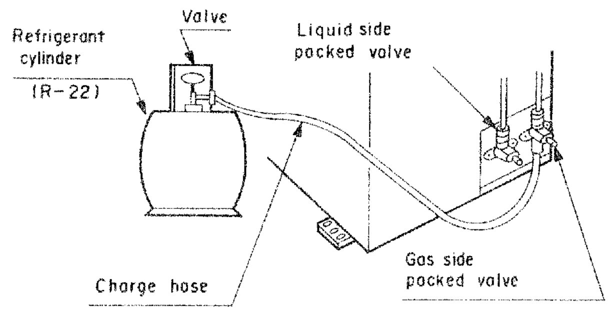
- Connect the charge hose (A) from the manifold valve to the charge inlet of the gas side packed valve.
- Connect the charge hose (B) to the port of vacuum pump.
- Open fully the low pressure side handle of the manifold valve.
- Drive the vacuum pump.
- Close the low pressure side handle of manifold valve after vacuumizing and stop the vacuum pump.



Continue vacuumizing more than 15 minutes and check the pressure gauge shows indicates – 760mm. Hg

Use of refrigerant cylinder

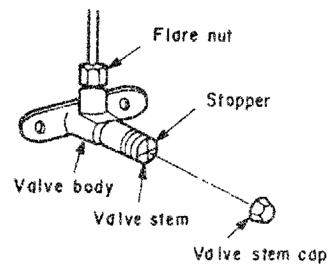
- Connect the charge hose from the refrigerant cylinder to the gas side packed valve charge inlet.
- Slightly loosen the flare nut of gas side packed valve.
- Open the refrigerant cylinder and supply refrigerant.
- Six or seven seconds after the gas begins to come; out from gas side packed valve, securely tighten the flare nut at gas side packed valve. (For proper tightening torque, see table at the "Tightening connection" section.)
- Open the stems of packed valves gas side and liquid side all the way.
- Securely tighten the stem cap to each of the packed valve stems.



Packed valve handling precautions

- Open the valve stem all the way out; do not try to open it beyond stopper.
- Securely tighten the valve stem cap with the wrench or the like.
- Valve stem cap tightening torque is as follows:

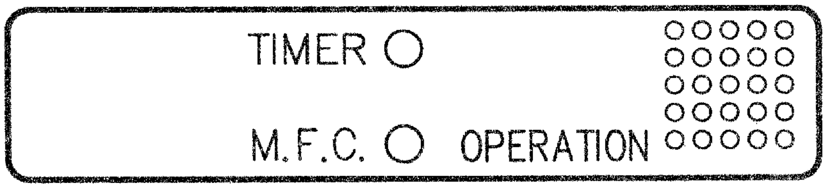
  Gas pipe (1/2 in. O.D.) side 500kg F-cm.
  Liquid pipe (1/4 in. O.D.)side 160kg F-cm.



9. OPERATION DESCRIPTION

NAMES AND FUNCTIONS OF INDICATORS AND CONTROLS ON INDOOR UNIT

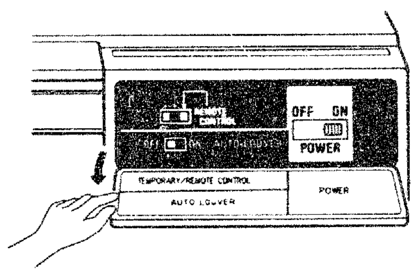
Display panel



- Operation indicator lamp .. Green LED: This lamp will light in green when unit is on.
- Timer indicator lamp .. Yellow LED: This lamp will light in yellow when timer is set.
- M.F.C. lamp .. Red LED: This lamp will light in red when mulfunction is checked.

(M.F.C.: Mal Function Check)

Control panel Close the panel cover after you set the control switches.

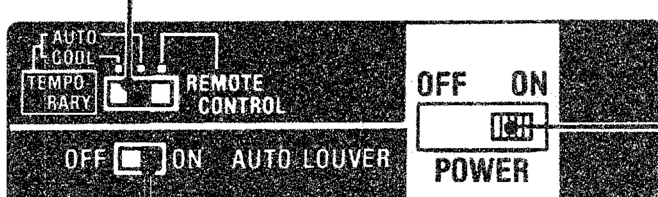


TEMPORARY operation switch

Normally set this switch to REMOTE CONTROL. If you misplace or lose the remote controller or its batteries are exhausted, set this switch to the AUTO position of TEMPORARY operation.

POWER switch

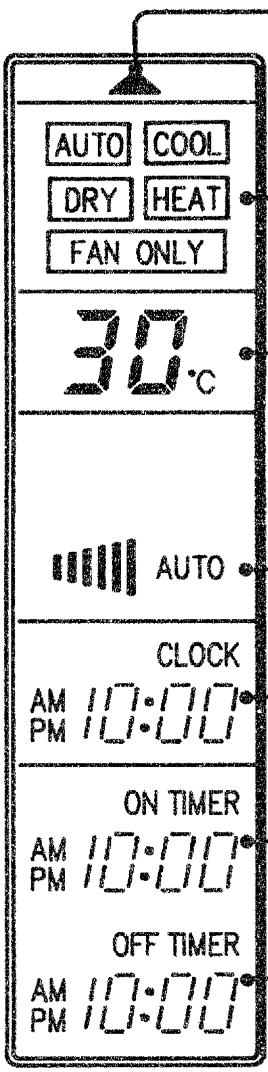
The main power ON/OFF switch controls both units of the air conditioner, normally set the switch to ON. Set it to OFF if you do not plan to use the unit for a few weeks.



AUTO LOUVER switch

Normally keep this switch in the ON position so the direction of the air flow will automatically change depending on the operation selected.

NAMES AND FUNCTIONS OF INDICATORS ON REMOTE CONTROLLER



Transmission Indicator

This transmission indicator lights when the remote controller transmits signals to the main unit.

MODE display

Indicate the current operation mode, framing it with a box ▭.

THERMO display*

Displays the temperature setting (17°C to 30°C). When you set the operating mode to AUTO or FAN ONLY, no temperature setting is displayed.

FAN SPEED display

Displays the selected fan speed. AUTO and three fan speed levels can be indicated. Displays AUTO when the operating mode is either AUTO or DRY.

CLOCK display

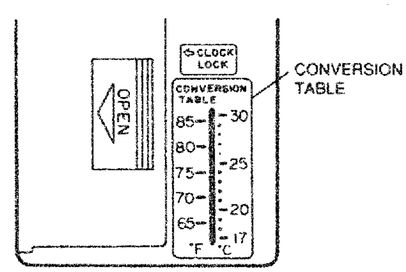
The current time is indicated.

TIMER display

The time set for timer operation is indicated.

- In the above illustration, all displays are indicated for the sake of clarity. During operation, only the relevant displays will be shown on the remote controller.

* See the CONVERSION TABLE back of the remote controller



REMOTE CONTROLS AND THEIR FUNCTIONS

Push the RESET/INDICATE button to display current operating settings.

Infrared signal emitter
Transmits a signal to the main air conditioner unit.

Remote control cover knob
Push this knob to the right to open the cover.

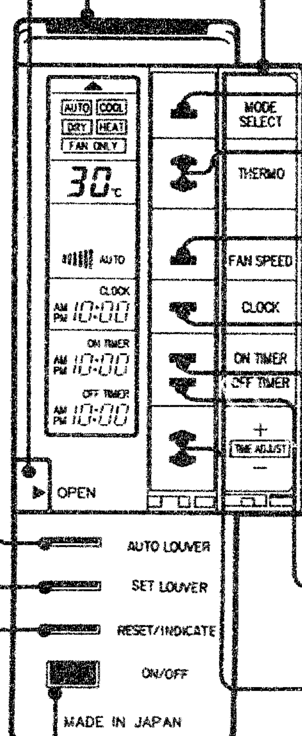
AUTO LOUVER button
Push this button to swing louver. (A receiving beep is heard.)
Push this button again to stop louver swinging. (A receiving beep is heard.)

SET LOUVER button
To adjust the air flow direction by the wireless auto louver during the air conditioner operation, push this button. (Receiving beep: Pe!)
While this button is kept depressed, the receiving tone continuously sounds.

RESET/INDICATE button
When a setting is changed during operation, its indicator on the remote controller display starts flashing. Push this button to set new condition and to stop the flashing. (A receiving beep is heard.)
If you push this button when the air conditioner is inoperative, the current settings are displayed and you can confirm their values. (No receiving beep is heard.)

ON/OFF button
Push the button to start operation. (A receiving beep is heard.)
Push the button again to stop operation. (A receiving beep is heard.)
If no receiving sound is heard from the air conditioner main unit, push the button twice.

Remote control cover



MODE SELECT button
Push this button to select modes. Each time you push the button, a mode is selected in a sequence that goes from AUTO, COOL, DRY, HEAT, FAN ONLY, and back to AUTO.

THERMO button
▲.... The set temperature is raised up to 30°C
▼.... The set temperature is lowered down to 17°C
(See the CONVERSION TABLE back of the remote controller.)

FAN SPEED button
Push this button to select fan speed. When you select AUTO, the fan speed is automatically adjusted according to the room temperature. You can also manually select the desired fan speed from three levels.

CLOCK button
Push this button to set the current time.

ON TIMER button
Push this button to set the ON timer.

OFF TIMER button
Push this button to set the OFF timer.

TIME ADJUST button
Push this button to set time.

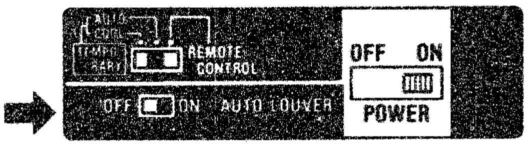
- The setting indicators on the remote controller display flash when a setting is changed. Push the RESET/INDICATE button to set the new conditions and stop the flashing.
- If you push the MODE SELECT button accidentally during operation and leave its indicator flashing, the operating mode will be changed to the one indicated when you turn on the unit next time.

ADJUSTING AIR FLOW DIRECTION

Adjustment of Vertical Air Flow Direction

When using the wireless auto louver

Set the AUTO LOUVER switch of the main unit control panel to ON.

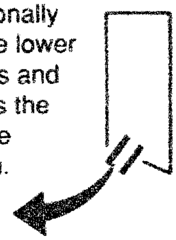


The air flow direction changes according to the operating condition.

During Heat Operation

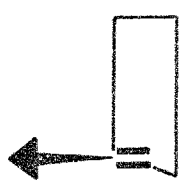
When the room temperature is low
The air flows downward and heats from floor level.

As the room temperature rises
The air flows diagonally to the lower levels and heats the whole room.

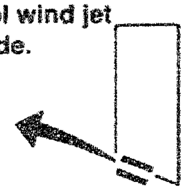


During Cool or Dry Operation

At the start of operation, the air flows horizontally and extends over the whole room.



When the room temperature reaches the set temperature, the unit enters the cool wind jet mode.

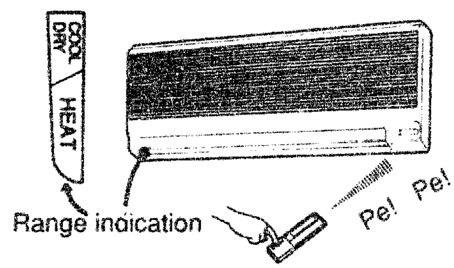


Adjusting the Louver Direction

Hold down the SET LOUVER button of the remote controller and set the air flow in the direction you disire.

During Cool, Dry and Heat Operations

Change the air flow direction within the range indicated by the side of the louver.



When you stop operation using the remote controller

The louver turns downward.

If you do not prefer the wireless auto louver operation

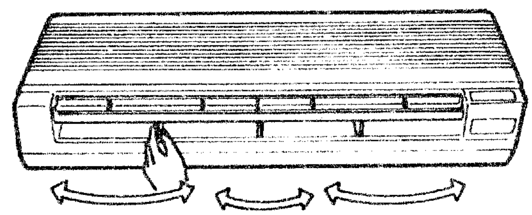
Set the AUTO LOUVER switch of the main unit control panel to OFF. During automatic operation, however, the air flow direction will change automatically according to the operation mode.

CAUTION

- To avoid condensation and damage to the louvers, do not use the downward air flow for a prolonged period when using the Cool or Dry mode.
- The air flow direction can be changed using the remote controller while the air conditioner is in use. When the air conditioner is turned on the next time, the air flow will automatically revert to horizontal during Cool or Dry operation or vartical during Heat operation.
- The air flow cannot be adjusted with the remote controller when the air conditioner is off or when the ON TIMER is setting.

Adjustment of Horizontal Air Flow Direction

- The horizontal air flow can be independently adjusted by means of the center, left and right grilles.
- Optionally adjust the horizontal air flow direction according to the room condition by moving each grille with a hand.

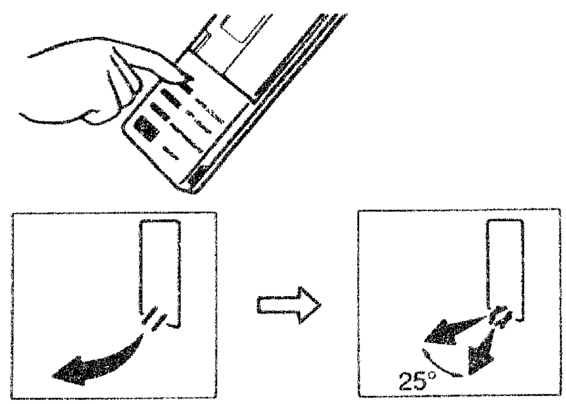


Horizontal air flow direction adjustment grilles

Other Function of Wireless Auto Louver

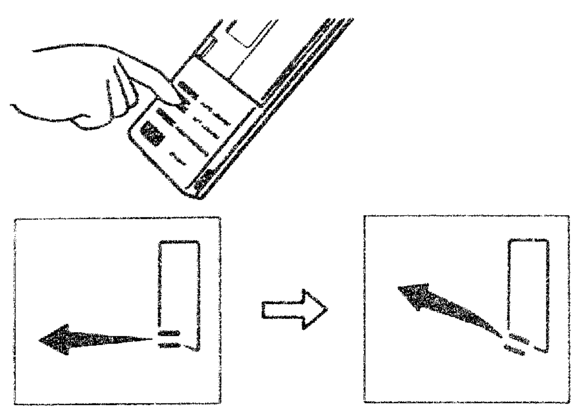
Louver Swing

- When the AUTO LOUVER button on the remote controller is pushed, the louver vertically swings within range of 25°.
  Remember to keep AUTO LOUVER button on the main unit in the ON position.
- To change the swing position, push the SET LOUVER button on the remote controller and then adjust the swing position.
- To stop the swing action, push the AUTO LOUVER button on the remote controller once again.



Cool Wind Jet

- If the SET LOUVER button on the remote controller is pushed while the air flows is horizontal, the louver direct the air flow upward. This will prevent a draft from blowing directly on your body.
  This option can be used with the COOL or DRY mode.
- To continue the cool wind jet feature, set the AUTO LOUVER switch to the "OFF" position after setting the louver in the desired position.
  You may use the louver swing option and the cool wind jet option at the same time.



Notes on handling the remote controller

- Do not throw and take care not to drop the remote controller. Do no allow water to spill on it.
- Do not place the remote controller in direct sunlight or near a stove.
- When the receiver (main unit) is subject to direct sunlight, the air conditioner may not operate normally. Shield sunlight using a curtain or any other suitable means.
- The signal may not be received in the room where a fluorescent lamp with an electronic glow starter is used. Place the conditioner as far away from the fluorescent lamp as possible.
- The air conditioner is not operative if there is an obstruction, such as a curtain, between the air conditioner and remote controller or if the air conditioner is far away from the remote controller. The distance within which the air conditioner can receive signal from the remote controller is about 7m. (This distance differs according to the battery strength, and the positional relationship between the remote controller and air conditioner.)
- The sensitivity of the receiver may be lowered if dust builds up on the transmitter (inferred rays emitter) or receiver (main unit).
- If another air conditioner or electrical apparatus is operated by the remote controller, place it away from the controller.
- This wireless remote controller is not interchangeable with those of previous business years.

OVERVIEW OF AIR CONDITIONER CONTROL

This air conditioner uses a TRIAC controlled variable-speed stepless induction motor as its indoor fan motor and a power proportionate control compressor which can change the motor speed in the range of 2100 to 7200r.p.m. to change the level of power. The indoor unit is equipped with a motor phase control circuit, and the outdoor unit has an inverter (frequency change device) for controlling the compressor motor.

The air conditioner as a whole is primarily controlled by the indoor unit control section.

The indoor control section drives the indoor fan motor according to an operation command signal from this remote controller and transfers the operation command signal (specification of cool operation and compressor operation frequency) to the outdoor unit control section. The outdoor unit control section receives the signal from the indoor unit control section controls the outdoor fan and four-way valve, controls the inverter output voltage and frequency supplied to the compressor motor, and changes the compressor motor speed. It also sends information concerning the operation condition of the outdoor unit back to the indoor unit control section.

1. Indoor fan control

Although the variable-speed stepless induction motor is used to drive the indoor fan, only four-speed stages are used in the AUTO position of fan speed and only three speed stages are used in MANUAL position for control purposes. No switching sound is heard when the speed is changed since no relay tap switching is used.

2. Capacity control

The cooling and heating capacity is varied by changing the compressor motor speed. The inverter converts the AC 220/240V for RAS-45BAHV into DC 310/340V to generate a 35 to 120Hz three-phase AC power by the transistor module (which contains six transistors) and feeds it to the compressor to change its speed. The range of frequencies fed to the compressor varies with different operation conditions. (cool or dry).

Cool operation:35 to 90Hz, Dry operation:35Hz Heat operation:35 to 120Hz

3. Current limit control

The outdoor unit control section (inverter assembly) detects input current to the outdoor unit. If the current exceeds the specified value due to compressor motor speed following the command from the indoor unit, the control section decreases the compressor motor speed to make it equal to or as close as possible but not greater than the speed specified by the command.

4. High-temperature limit control (Antifreeze control)

When the refrigerant condensation temperature detected by the indoor heat-exchanger sensor during the heat operation (refrigerant evaporation temperature during cool or dry operation) is equal to or more (less) than the specified value, the compressor speed is reduced and operated at a temperature below (over) the specified temperature to prevent overpressure (indoor heat exchanger freezing).

5. Louver control

When the air conditioner is operated with the LOUVER button pressed ON, the louver position is automatically controlled. The louver swings if the AUTO LOUVER button on the remote controller is pressed. The louver position can be set freely by pushing the SET LOUVER button.

6. Thermal radiation sensor control

The set temperature is adjusted acoording to the radiant temperature. The adjustment temperature varies with different operation mode (cool, dry, heat).

OPERATION CIRCUIT DESCRIPTION

1. When the power cords are connected to AC 220, 240V (50Hz) outlet for 45BAHV, and the POWER SUPPLY switch is set to ON the operation (ON) lamp starts flashing, indicating that power has been turned on (or that power failure occurred).

2. Confirm that the remote control TEMPORARY switch is set to the REMOTE CONTROL position.

3. When the remote control ON/OFF button is pushed, a receive tone is heard from the main unit and the following operations are carried out.

   (1) Fan operation (the remote control MODE button is set to the FAN ONLY position).

   - Once the setting is made, the operation mode is memorized in the microcomputer so that the same operation can be effected thereafter simply by pushing the ON/OFF button.
   - When the operation mode was changed, the mode indicator starts flickering. So never fail to push the RESET/INDICATE button.
   - When the FAN SPEED button is set to the AUTO position, the indoor fan motor operates as shown in Fig. 1. When the FAN SPEED button is set to LOW, MED, or HIGH, the motor operates with a constant air volume as listed in Table 1.
   - When the room temperature lowers, the air volumes if reduced even during operation with a high airflow. Power relay K8 is turned off.

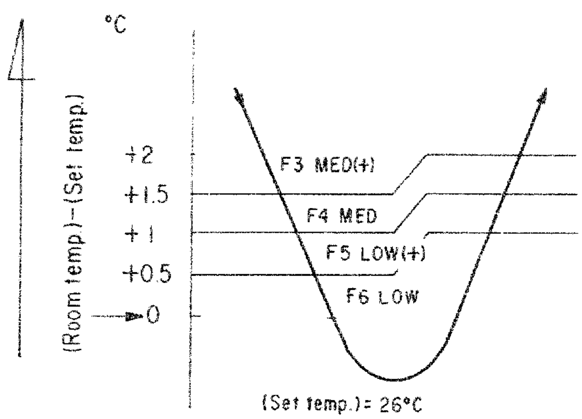


Fig. 1 Auto setting of air volume

Indication of FAN SPEED	Air volume
▮▮	F6 LOW
——	F5 LOW (+)
▮▮▮▮	F4 MED
——	F3 MED (+)
▮▮▮▮▮▮	F1, F2 HIGH

Table 1 Manual setting of FAN SPEED

(2) Cool operation (The remote control MODE button is set to the COOL position)

- Once the setting is made, the operation mode is memorized in the microcomputer so that the same operation can be effected thereafter simply by pushing the ON/OFF button.
- When the operation mode is changed, the remote controller's indicator lamp starts flickering. So never fail to push the RESET/INDICATE button.
- Power relay K8 is turned on to energize the outdoor unit, and a cool operation signal is transmitted.
- The indoor fan motor operates as shown in Fig. 2 when FAN SPEED button is set to AUTO.
- The motor operates with a constant air volume as listed in Table 1 in previous page, when the FAN SPEED button is set to LOW, MED, or HIGH.
- It is to be noted that the air volumes is reduced when the room temperature lowers even during operation with high volume.
- The outdoor unit controls the outdoor fan relay K1 and the compressor motor speed according to the operation command signal sent from the indoor unit. Both the four-way valve relay K5 and the two-way valve relay K6 are always OFF.

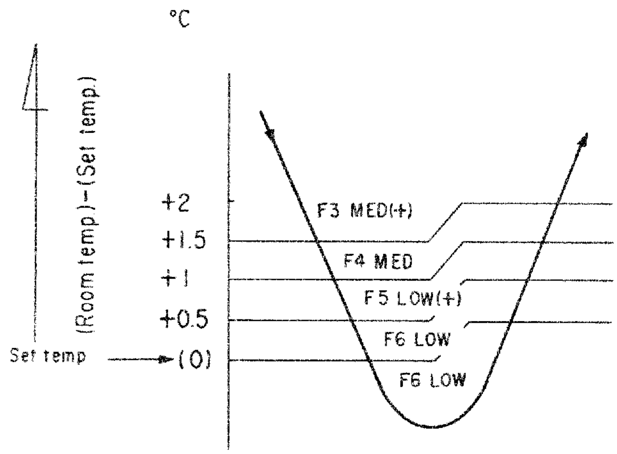
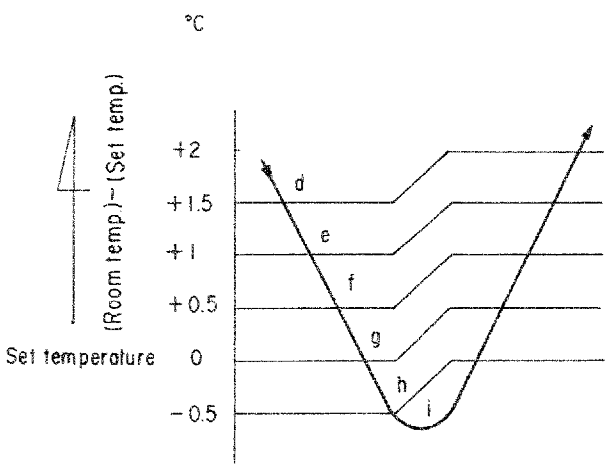


Fig. 2 AUTO

1) Cool capacity control

- The cool capacity and room temperature are controlled by changing the compressor motor speed according to both the difference between the temperature detected by the room temperature sensor and the temperature set by THERMO button and also any change in room temperature as shown in Fig. 3.
- The 60Hz operation is performed for only two minutes when the compressor is started and restarted.



Compressor operation frequency at start

Zone	Frequency
d S9	90Hz
e S7	80Hz
f S5	43Hz
g S3	35Hz
h, i	0Hz (Compressor OFF)

Fig. 3

NOTE:

1. When the room temperature is in the same zone for three minutes, change the operation frequency.
   d to e: +10 to 15Hz
   f : no change
   g : –10 to 15Hz
   h : off
2. When the zone is changed, the operation frequency is increased or decreased by 10 to 15Hz.
3. The maximum frequency is 80Hz when a 90Hz operation is performed for 29 minutes and more at a room temperature of less than 29°C.

2) Antifreeze control
- The indoor heat-exchanger sensor detects the indoor heat-exchanger temperature. When the temperature drops to 0°C or lower, the compressor speed is decreased gradually to prevent the indoor heat-exchanger from freezing. When the temperature reaches or exceeds 2°C, the capacity is controlled as described in 1).

3) Current limit control
- The outdoor current sensor detects input current to the compressor (namely, the inverter main circuit section), which utilizes most of the input to the air conditioner. When the current value exceeds 13.5A (at 220V), or 12.0A (at 240V), the compressor speed is reduced gradually to maintain the value at 13.5A (at 220V), or 12.0A (at 240V) or less. When the current decreases to about 14.0A (at 220V) or 12.5A (at 240V), the capacity is controlled as described in 3-(2)-1) and as shown in Fig. 3.

4) Louver control
- When the LOUVER button is pushed during the operation, the louver automatically selects almost horizontal position.

(3) Dry operation (The remote control MODE button is set to the DRY position.)
- Once the setting is made, the operation mode is memorized in the microcomputer so that the same operation can be effected thereafter simply by pushing the ON/OFF button.
- When the operation mode is changed, never fail to push the RESET/INDICATE button as the remote controller's indicator lamp starts flickering.
- Power relay K8 is turned on to energize the outdoor unit. Either moderately cool dry or fan operation is selected according to the difference between the set temperature and the room temperature, and an operation signal is transmitted.

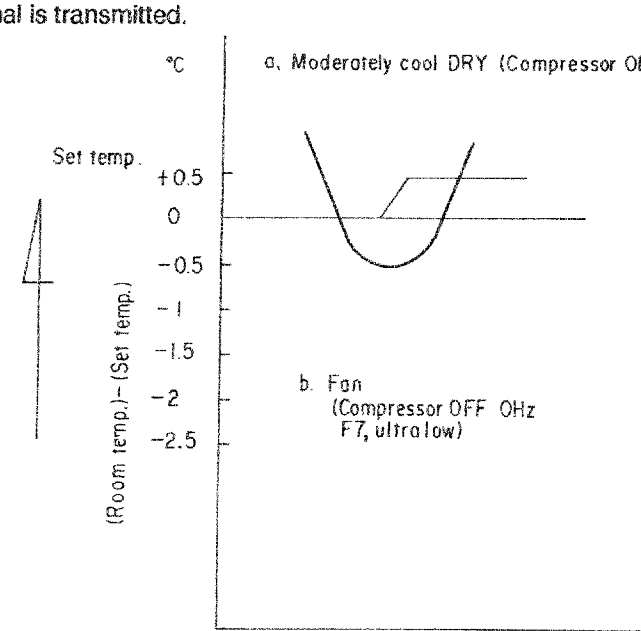


Fig. 4

1) At moderately cool DRY operation
  - If room temperature is higher than set temperature, indoor fan motor is operated at ultra-low as shown in Fig. 4, while outdoor unit controls outdoor fan relay K1 in accordance with operation signals sent from the indoor unit and also performs control on the compressor motor speed.

2) At FAN ONLY operation
  - When room temperature is lower than set temperature, indoor fan motor is operated at ultra-low as shown in Fig. 4 while the outdoor unit stops.

(4) HEAT operation (The remote control MODE button is set to the HEAT position)
- Power relay K8 is turned on to energize the outdoor unit and a heat operation signal is transmitted.
- When the SELECTOR knob is set to AUTO, the indoor fan motor operates as shown in Fig. 5, and when the knob is set to LOW, MED, or HIGH, the motor operates with a constant air volume.
- When the room temperature rises, the volume of strong wind is decreased.
- The outdoor unit, according to the operation signal sent from the indoor unit, controls the outdoor fan relay K1 and K2.
- The four-way valve relay K5 is always ON and two-way valve relay K6 is always OFF during operations other than defrost operation.

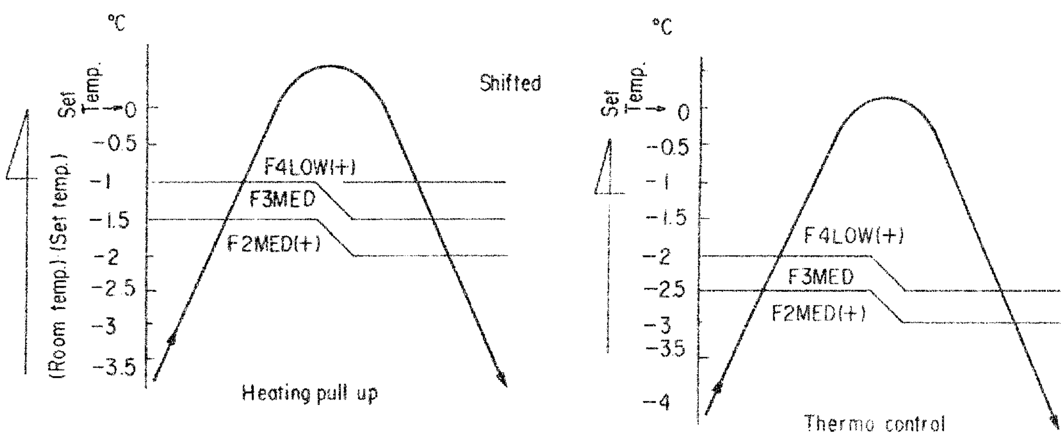


Fig. 5

1) Heating capacity control
  - The heating capacity is controlled by changing the compressor speed as shown in Fig. 6 according to both the difference between the temperature detected by the room temperature sensor and the temperature setting made by the THERMO button and also the change in room temperature.
  - The 60 Hz operation is performed for only two minutes and the 90Hz operation is performed for 5 minutes when the compressor is started and restarted.

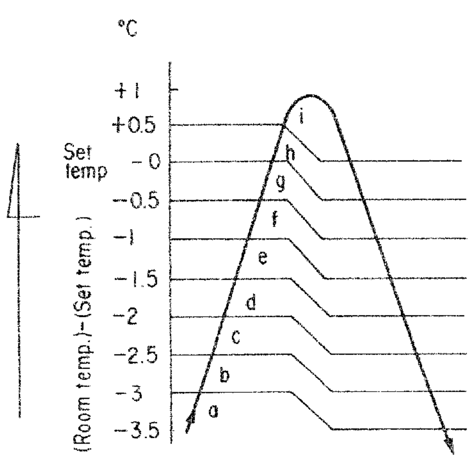


Fig. 6

Compressor operation frequency at start

Zone	Frequency
a	120Hz
b	100Hz
c	85Hz
d	75Hz
e	60Hz
f	50Hz
g	35Hz
h, i	0Hz (Compressor OFF)

NOTE:

1 When the room temperature is in the same zone for three minutes, change the operation frequency.
 a to e : +10 to 15Hz
 f : no change
 g : − 10 to 15Hz
 h : off

2. When the zone is changed, the operation frequency is increased or decreased by 10 to 15Hz.

3. The maximum frequency is 90Hz when an operation is performed at 120Hz or more for 29 minutes and more at a minimum room temperature of 20°C and with a minimum heat-exchanger temperature of 38°C.

2) High-temperature limit control
- When the indoor heat-exchanger temperature detected by the indoor heat-exchanger sensor is 55°C or above, the compressor speed is reduced gradually to prevent excessive increase of condensation pressure.
- If the temperature drops below 53°C , the control is performed as described in 1) in previous page.

3) Cool air flow control
- If the indoor heat-exchanger temperature detected by the indoor heat-exchanger sensor is 20°C or below, the indoor fan stops, and if the temperature rises to 32°C or above, the fan is restarted.

4) Current limit control
- The outdoor current sensor detects input current to the compressor as in the cool operation. The compressor speed is controlled to maintain a current of 16.5A (at 220V) or 15.0A (at 240V) or less.
- When the current drops to 15.0A (at 220V) or 13.5A (at 240V), the capacity is controlled as shown in Fig. 5.

5) Defrost operation
- The time of the compressor operations accumulated, the defrost operation is performed after 47 minutes of operation (11 minutes when the power is turned on just once), and if the outdoor heat exchanger temperature detected by the outdoor heat exchanger sensor is 8°C or less for 2 minutes.
- At this time, relays K1 and K2 are turned off and K6 is turned on. So the outdoor fan is turned off and the two-way valve are turned on.
- The indoor fan is turned off by the cool air flow control described in section 3) above.
- The defrost operation stops and the heat operation restarts when any one of the following conditions is satisfied:
- The outdoor heat-exchanger temperature rises 5°C or more.
- Defrost operation continues for maximum10 minutes.

6) Louver control

7) Thermal radiation sensor control

(5) TIMER operation
- Once the setting is made, the operation is memorized in the microcomputer so that the same operation can be effected thereafter simply by pushing the ON/OFF button.

To stop the operation of the air conditioner:
- Push the ON/OFF button.

To change the preset time of the timer halfway:
- Push the TIMER button to set the desired time and then push the RESET/INDICATE button. The timer is activated when the reset time has passed after the point when the RESET/INDICATE button was pushed.

To make the operation continuous halfway in the course of the timer operation:
- From OFF to CONTINUE
  Push the TIMER SELECTOR button to get into CONTINUE and then push RESET/INDICATE button.
- From ON to CONTINUE
  Push the TIMER SELECTOR button to get into CONTINUE and then push the RESET/INDICATE button.

10. TROUBLESHOOTING CHART

The indoor and outdoor units both incorporate electronic controllers.
Check the units by the following procedures.
(The check points listed on the circuit diagrams affixed to the inside of the indoor and outdoor units will be helpful for service.)

CAUTION

1. A large capacity electrolytic capacitor is used in the outdoor unit controller (inverter). It takes time for the charge in the capacitor to discharge after power is turned off. (Some twenty minutes are required for the charge to dissipate completely.) Remember that touching a live part before the capacitor discharges will result in electric shock.
2. The inverter is designed to be repaired at a replaceable block level.

1. Primary Checks

1-1. Checking the power supply voltage

Make sure the power supply voltage is AC 220/240V for 45BKHV(W) ±20%. The units may not operate correctly if the voltage is wrong.

1-2. Checking cable connections between the indoor and outdoor units

Five cables connect the indoor and outdoor units. Make sure each pin of the indoor terminal block is connected to its counterpart of the outdoor terminal block that has the same number. The outdoor unit does not operate if connection is wrong. Note that the equipment will not be damaged even if a wrong connection is made.

1-3. Misleading but good operations (program controlled operations)

The symptoms listed in table 1 (next page) appears as the air conditioner inbuilt microprocessor operates under the control of programs. If a user complains about one of the symptoms, explain to the person that the device is not malfunctioning but is designed to operate that way for reasons of control. Check if a user's complaint is against such a symptom.

Table 1 Misleading but good operations

No.	Operation of Air Conditioner	Description
1	Indoor unit's ON lamp flashes when indoor unit's POWER SUPPLY switch is turned on.	ON lamp flashes to indicate application of power. Flashing stops once ON/OFF button is operated. (Flashing occurs also when electric current is shut off.)
2	Compressor does not work though room temperature is in the range of turning the compressor on.	Compressor does not work while the compressor restart delay (3-min.) timer is active. The same is true after power is turned on, as the time is still active.
3	In dry operation, there is not change in air volume even if operating FAN SPEED button.	Indication of air volume is fixed to AUTO. In dry operation, it is fixed to "ultra low".
4	Compressor operating frequency stays at 60Hz when it restarts.	When the compressor restarts after stopping due to the setting of room temperature, the operating frequency is limited to 60Hz for a minute after the restarting.
5	When ON/OFF button has been pushed to stop operation, power relay of indoor controller turns off after a delay of 2-3 seconds.	Time is needed for the outdoor unit controller to execute the stop sequence of the compressor, outdoor fan, etc. after receiving a stop signal from the indoor unit controller. For this reason, a 2-3 sec. delay is incorporated before operating the power relay which sends power from indoor to outdoor unit.
6	Compressor operating frequency stops rising about 30 sec. after start of operation and it begins to rise after additional 30 sec. or so.	The compressor operating frequency is limited to 60Hz within two minutes and 90Hz within five minutes after the start of operation so that compressor operates smoothly. After two or five minutes, the frequency begins to rise to the specified level.
7	Louver does not operate when operation starts with LOUVER switch is turned on.	If the operation status is the previous one in which the LOUVER button for remote control was not operated, the louver position is not set again.
8	During the counting of 1-minute timer the compressor does not turn off when starting and restarting the operation even under the condition of turning off the compressor.	While 1-minute timer is counting, data of thermo sensor is not read. If the setting of temperature is changed, however, the 1-minute timer is short-circuited.
9	During the heating operation, compressor operating frequency stops rising to 120Hz, while the room doesn't achieve the setting temperature.	This control is performed automatically when the heat exchanger temperature exceeds a certain value so that it is decreased to prevent overload to the supply.

2. Roughly locating the source of trouble

2-1. Roles of the controllers

The controllers of the indoor and outdoor units perform different roles. By understanding the differences the source of trouble in either the indoor or outdoor unit can be easily located.

(1)Roles of the indoor unit controller

It decodes operation commands given by remote controller and performs the following functions:
Receiving signals from the room temperature (thermo) sensor and thereby measuring the air temperature at the inlet to the indoor heat exchanger.

- Receiving signals from the heat exchanger sensor for setting temperature of indoor heat exchanger (anti-freeze control, cool air flow control and High-temperature Limit control).
- Control of the louver motor.
- Control of the indoor unit fan motor.
- Control of the LED indicators.
- Controlling the power relay which turns on and off power sent to the outdoor unit, and sending operation command signals (serial signals) to the outdoor unit.
- Receiving operating status signals (serial signals) from the outdoor unit, detecting faulty conditions, and displaying such conditions.

(2)Roles of the outdoor unit controller

Receiving power (AC 220/240V for RAS-45BKHV(W)) and operation command signals from the indoor unit, the outdoor unit controller performs the following functions:

- Control of the compressor motor.
- Control of the outdoor fan motor.
- Control of the four-way valve.
- Control of the two-way valve.

According to the serial signals coming from the indoor unit.

- Detecting inverter input current and current limit operation.
- Overcurrent detection and protection of the transistor module (stopping the compressor motor).
- Defrost control during heat operation (temperature monitoring by outdoor heat exchanger sensor and control for both two-way valve and outdoor unit fan motor).
- Stopping the compressor motor and outdoor unit fan motor when serial signals do not reach the outdoor unit control circuit board assembly due to failure of signal circuits.
- Sending signals (serial signals) indicating operating status of the outdoor unit.

(3)Operation command signals (serial signals) sent from the indoor unit controller to the outdoor unit controller

The indoor unit controller sends the following two kinds of signals from the indoor unit controller:

- Identification of cool, heat or dry operation mode set by remote controller.
- Command signal of the compressor's operating frequency that depends on the room and preset temperature (together with corrections made according to change of the room temperature and the indoor heat exchanger temperature).
- While monitoring current input to the inverter, the outdoor unit controller follows up the above-mentioned signals (cool/heat/dry and the compressor's operating frequency signals) to the extent that the current supplied to the inverter is held at a permissible level.

(4)Operation status signals (serial signals) sent from the outdoor unit controller to the indoor unit controller

The outdoor unit controller sends the following signals to the indoor unit controller:

- Identification of the current operating mode
- Current operating frequency of the compressor
- Whether or not the protection circuit is working

The indoor unit controller monitors these signals and determines presence or absence of fault in terms of the following:

- Whether the current operation status is consistent with the operation command signals.
- Whether the current compressor operating frequency is consistent with the operation command signals.
- Whether protective circuit is working.

When no signals come from the outdoor unit, the indoor unit controller judge the status of the outdoor unit as faulty.

2-2. Methods of roughly locating the source of trouble

Which is the source of trouble, indoor or outdoor, can be determined primarily by the below-mentioned two methods:

$\boxed{\text{Judging from flashing of the indoor unit indicators}}$

Monitoring the operation status of air conditioner, the indoor unit indicates the self-diagnosis on the indoor indicators at replaceable block level as follows when the protective circuit works:

Item	Indication at Block Level	Contents of Self-Diagnosis
A	OPERATION lamp flashing (1Hz)	Power failure (when power is turned on).
B	OPERATION lamp flashing (5Hz)	Indoor P.C. board's protective circuit worked.
C	OPERATION/TIMER lamps flashing (5Hz)	Connective cable connection and serial signal block's protective circuit worked.
D	OPERATIONAL/M.F.C. lamps flashing (5Hz)	Outdoor P.C. board's protective circuit worked.
E	All the lamps flashing (5Hz)	Protective circuits for others (including compressor) worked.

Note:

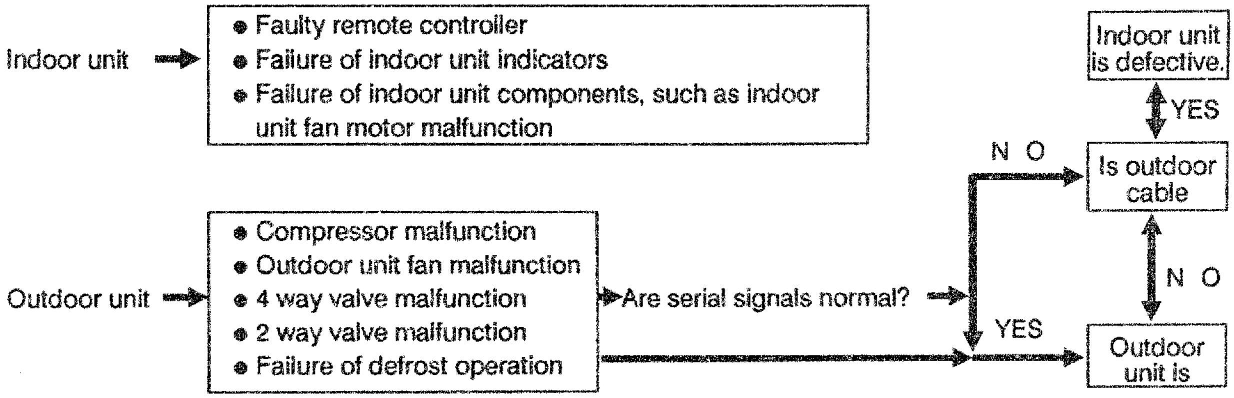
(1)The contents of the above items B, C and part of E are indicated when air conditioner is in operation.

(2)In case the items B and C, and B and part of E occurs concurrently, the block indication of item B will take priority.

$\boxed{\text{Judging from a faulty condition or component}}$

2-2-1. Self-diagnosis with remote controller

With the indoor unit controller, self-diagnosis of protective circuit action can be done by turning the remote controller operation into service mode, operating the remote controller, observing the remote controller indicators and checking whether ON lamp flashes (5Hz).



2-2-2. How to select remote controller operation mode

(1)Selecting service mode

Set selector switch beside the battery case of wireless remote controller to the service mode as shown in the figure below and short-circuit ACL (All Clear) terminals. Check 00 is indicated on the temperature display section and other indicators go off.

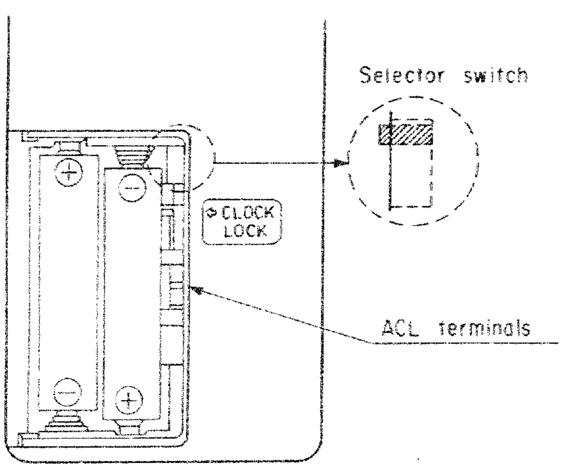
(2)Selecting ordinary mode

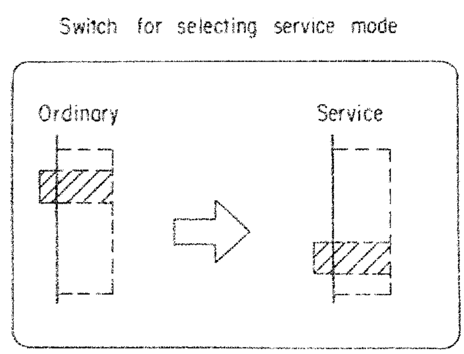
Set the above switch to ordinary mode as illustrated below, and short-circuit ACL (All Clear) terminals. At this point, make sure that all the indicators go off.

2-2-3. Cautions when doing service

(1)After finishing service, never fail to set the selector switch to the ordinary mode and short-circuit ACL(ALL Clear) terminals to get back into the ordinary mode.

(2)Normally, do not operate other switches than the service mode.





2-2-4. Self-diagnosis by check code

(1)Self-diagnosis by check code is carried out while the B-E block indication is being made.

(2)When switched to the service mode, remote controller key operation is done with "ON/OFF", "RESET/INDICATE" and "SET LOUVER". In doing this, operation of the keys gives indications on the remote controller as follows. (2-digit numerals are represented by hexadecimal.)

Operating key	Indication after operation
ON/OFF	00
SET LOUVER	+1 to the data before operation (Example) 02 - 03
RESET/INDICATE	−1 from the data before operation (Example) 02 - 01

(3)The following is the procedure for making self-diagnosis by check code:

1) Turn the remote controller into service mode and check that the remote controller's temperature indicator shows " 00 ".
2) Operate "ON/OFF" key and make sure that the timer indicator starts flashing (5Hz). During the self-diagnosis by check code, the timer indicator always flashes.
3) In this event, if the ON lamp flashes also, it can be known that protection circuit of indoor PC board is involved.

   If the ON lamp does not flash, you may confirm that the indoor PC board protective circuit is not in action.
4) As you operate "SET LOUVER" key, the remote control indication shows " 01 " for checking whether the ON lamp flashes or not. If the ON lamp flashes, you can confirm that the action of protection circuit for connective cable connection /serial signals is in action.
5) Get the +1 indications successively by operating "SET LOUVER" key for checking the contents of the self-diagnosis as shown in the table below. It is to be noted that " 00 " to " 03 " are for checking the presence or absence of the action of protection circuit for each block, while " 04 " to " 1F " are for checking whether typical protective circuit actions are involved in the symptom.

Block level		Diagnosis function				Judgment and action
Check code	Block	Check code	Symptom	Air conditioner status	Condition	
0 0	Indoor PC board	0 C	Thermo sensor short/break.	Continued operation	Indicated when detected abnormal.	1. Check thermo sensor. 2. If it is OK, check PC board.
		0 d	Thermo sensor off/break.	Continued operation	Indicated when detected abnormal.	1. Check heat exchanger sensor. 2. If it is OK, check PC board.
	No indication	0 F	Thermal radiation sensor off/break.	Continued operation	Indicated when detected abnormal.	1. Check thermal radiation sensor. 2. If it is OK, check PC Board.
0 1	Cable connection/serial signals	0 4	No return serial signal comes from start. 1) Wrong wiring of connective cable 2) Compressor thermal relay. Gas empty or leaks.	Continued operation	Reset to clear abnormal flashing of return serial signal.	1. If outdoor unit is totally dead: 1) Check flat cable correct if wiring is wrong. 2) Check 25A fuse of outdoor unit. 2. If indicated for "OTHER" block in operation, thermal relay for compressor. Fill up gas (check gas leak also). 3. When operating normally during check. If there is return signal between indoor-terminal block 4-5, replace inverter PC board. If no such signal replace indoor PC board.
			No operation command signal comes to outdoor unit.	Continued operation	Reset to clear abnormal flashing of operation command signal.	No operation command signal between indoor terminal block, 4 and 5. – Replace indoor PC board. If there is such signal: – Replace inverter PC board.

Block level		Diagnosis function				Judgment and action
Check code	Block	Check code	Symptom	Air conditioner status	Condition	
02	Outdoor PC board	13	1) Error in cool ing signal. 2) Trouble in outdoor inverter.	All off	Indicated when detected abnormal.	Replace PC board of inverter.
		14	Action of protection circuit for inverter over-current.	All off	Indicated when detected abnormal.	If return, totally stops right away. Check giant transistor's conductivity and base drive output of inverter PC board in a pair. If not good, replace the parts.
		18	During the heating opera tion, outdoor heat exchanger sensor off/break/short	All off	Indicated when detected abnormal.	Replace PC board of inverter.
03	Others (includ. compressor)	07	Return serial signals come at start but ceased to come halfway. Compressor thermal relay. Gas empty.	Continued operation	Reset to clear abnormal flashing of return serial signal.	1. ON/OFF repeated at an internal of about 10 to 40 minutes (not indicated when operating). – Fill up gas (Also check for gas leaks). 2. Air conditioner runs nor mally during checking. No return serial signal be tween indoor terminal block 4 and 5: – Replace inverter PC board. If there is such serial signal: – Replace indoor PC board.
		1d	Compressor does not run (A certain time elapsed after start of compressor, overcurrent protective circuit worked.)	All off	Indicated when detected abnormal.	1. Trouble with compressor. 2. Wrong wiring in compressor (lack of phase).
		1F	Compressor breakdown.	All off	Indicated when detected abnormal.	Check AC voltage (208-230V).

2-3 Judging from faulty condition or component

Table 3

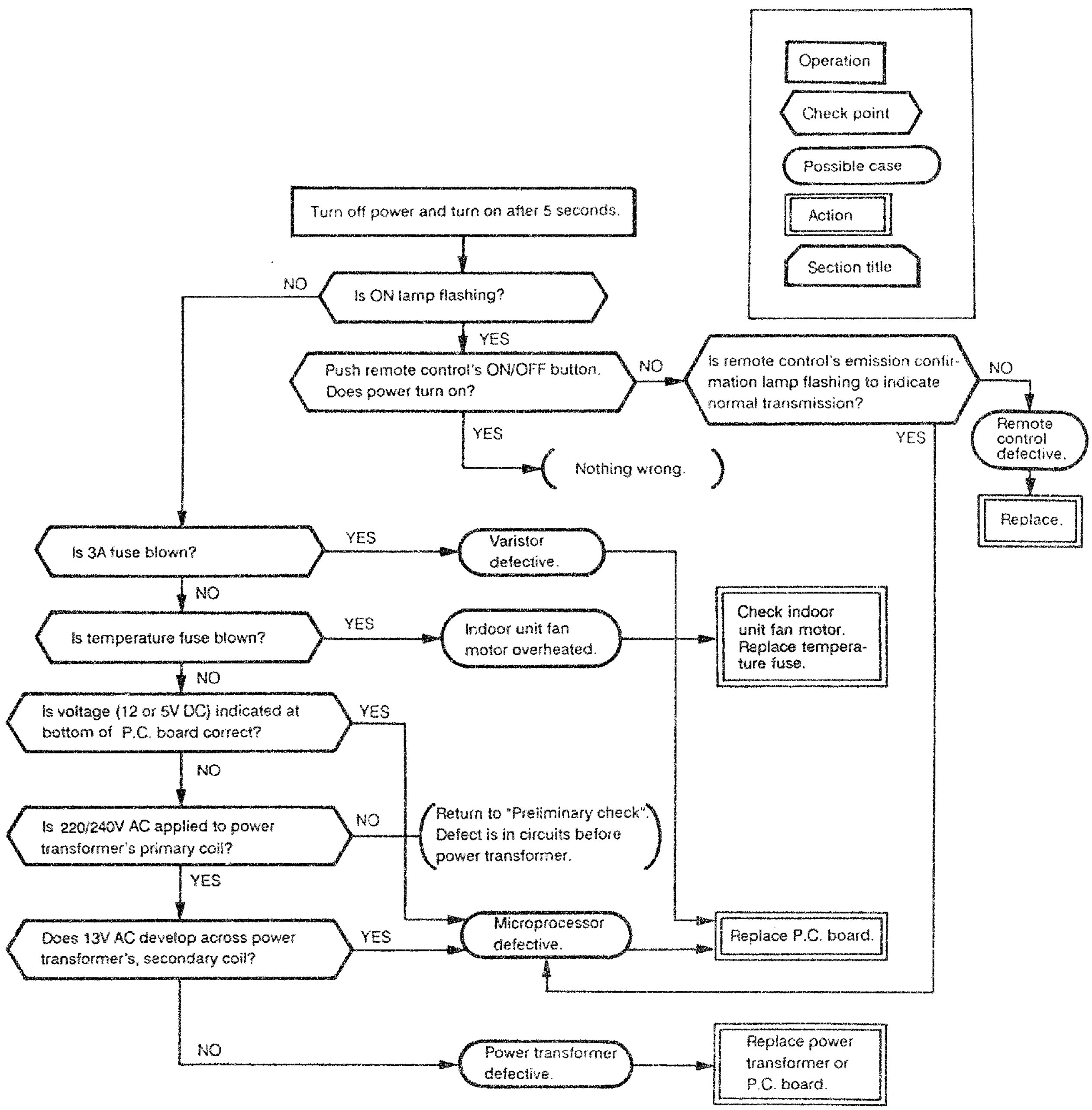
No.	Symptom	Action & result		Judgment
1	Remote control disable.	Turn off power, then turn on and operate by remote controller.	1) Remote control still disable.	Indoor unit (including remote control) defective.
			2) Remote control enable.	Good
2	Indoor unit fan does not	Does not operate in cooling and fan operation also.		Indoor unit defective.
3	Outdoor fan does not operate.	(1) Compressor operates.		Outdoor unit defective (including defect of outdoor fan motor).
		(2) Compressor does not operate, too.		Indoor unit defective.
4	Defrost operation faulty.			Outdoor unit defective (or gas leakage).

3. Troubleshooting flowcharts

3-1. Power does not turned on (air conditioner is completely dead).

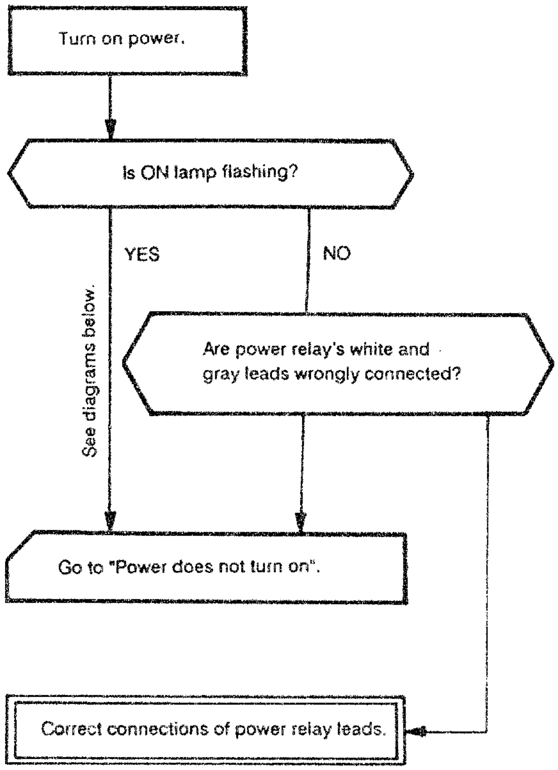
<Preliminary checks>

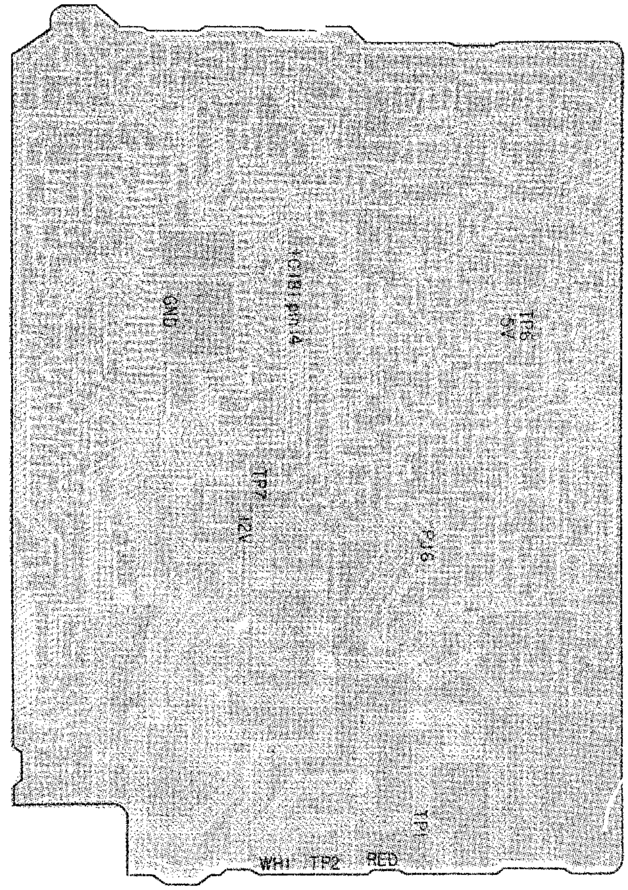
(1)Is the power voltage correct?
(2)Are cable connected properly?
(3)Does the POWER SUPPLY switch set to ON?
(4)Are the power transformer's primary and secondary circuit connectors conncted to the printed circuit board?

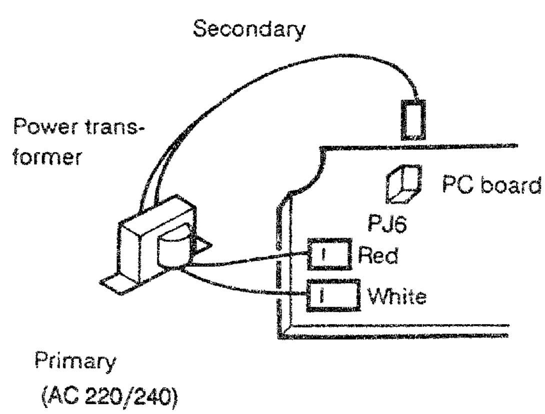
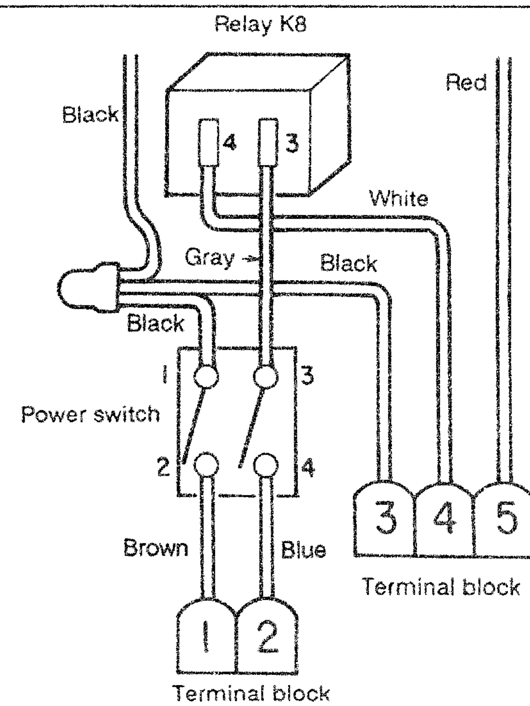


3-2. Power does not turn on after PC board replacement in indoor unit.

<Checking procedure>





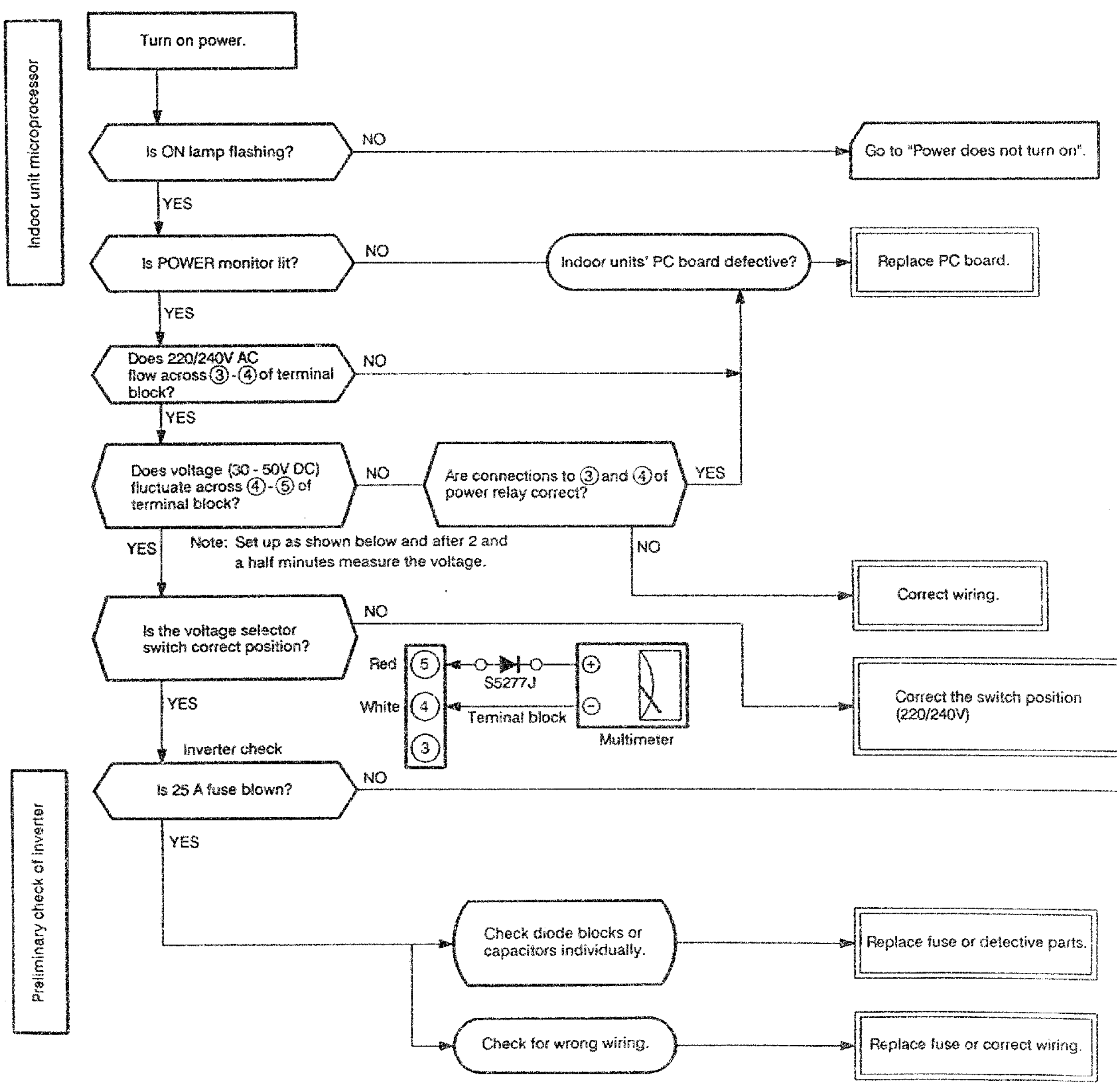
Power transformer	Power relay
	

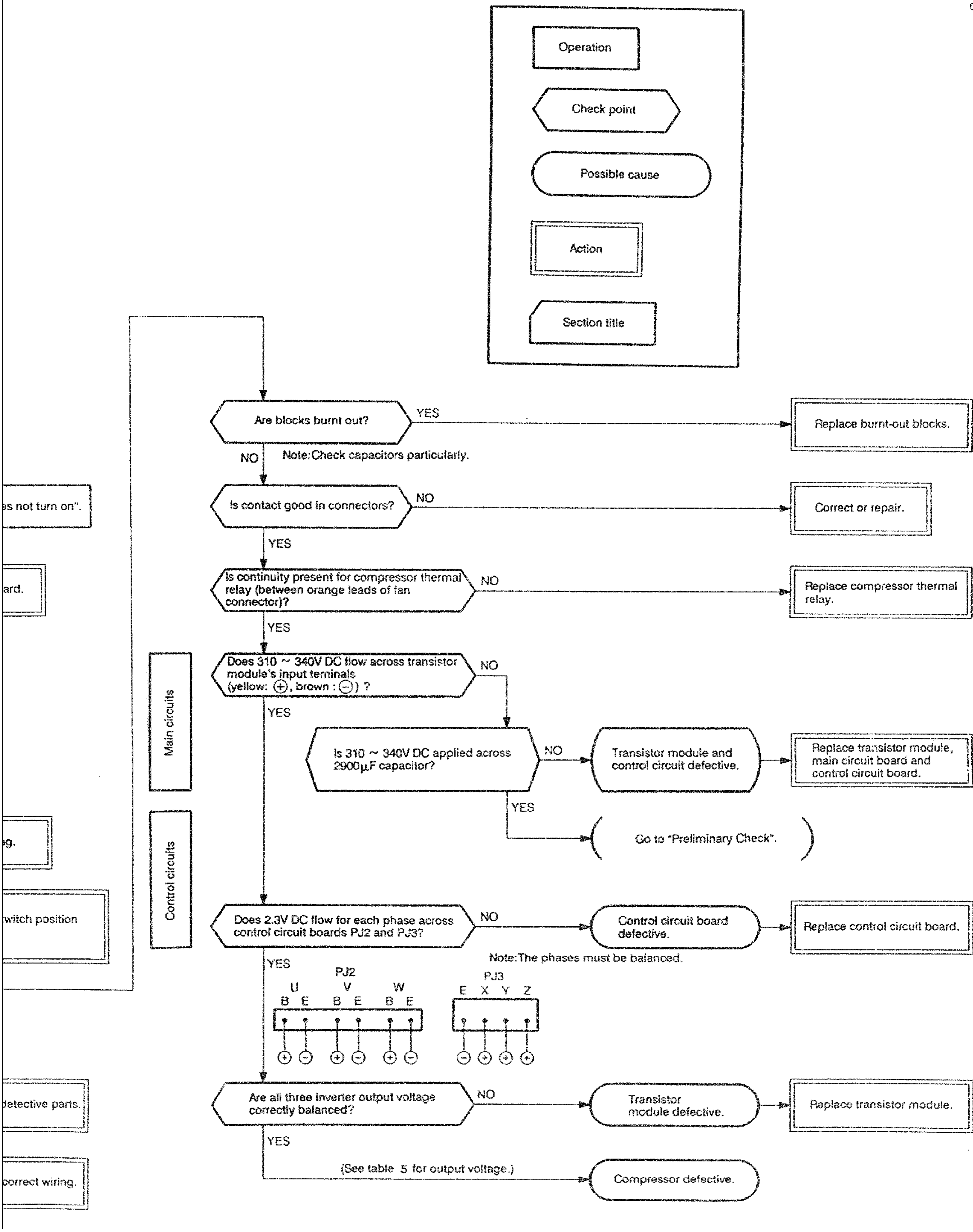
3-3. Compressor does not operate (and outdoor unit fan as well).

<Preliminary check>

(1)Is the temprature set by remote control higher than the room temperature in the cool operation?
(2)Is the temperature set by remote control lower than the room temperature in the heat operation?
(3)Is FAN ONLY mode selected by remote controller?
(4)Are cables connected properly?

<Checking procedure>



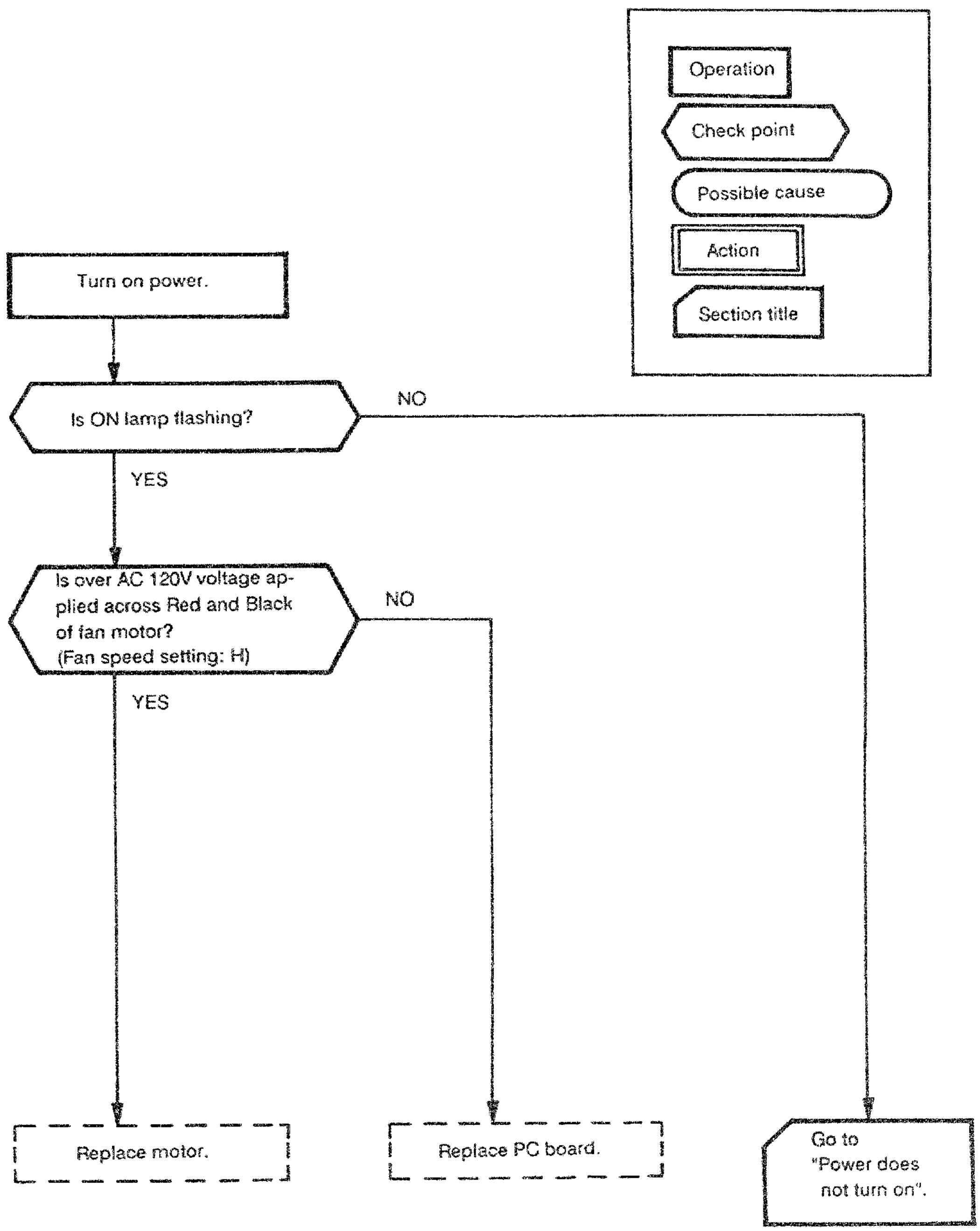

Operation
Check point
Possible cause
Action
Section title
Are blocks burnt out?
YES
Replace burnt-out blocks.
NO
Note:Check capacitors particularly.
Is contact good in connectors?
NO
Correct or repair.
YES
Is continuity present for compressor thermal relay (between orange leads of fan connector)?
NO
Replace compressor thermal relay.
YES
Main circuits
Does 310 ~ 340V DC flow across transistor module's input terminals (yellow: ⊕, brown : ⊖) ?
NO
YES
Is 310 ~ 340V DC applied across 2900μF capacitor?
NO
Transistor module and control circuit defective.
Replace transistor module, main circuit board and control circuit board.
YES
Go to "Preliminary Check".
Control circuits
Does 2.3V DC flow for each phase across control circuit boards PJ2 and PJ3?
NO
Control circuit board defective.
Replace control circuit board.
Note:The phases must be balanced.
YES
PJ2
U
V
W
B E B E B E
⊕ ⊖ ⊕ ⊖ ⊕ ⊖
PJ3
E X Y Z
⊖ ⊕ ⊕ ⊕
Are all three inverter output voltage correctly balanced?
NO
Transistor module defective.
Replace transistor module.
YES
(See table 5 for output voltage.)
Compressor defective.

3-4. Indoor unit fan alone does not operate.

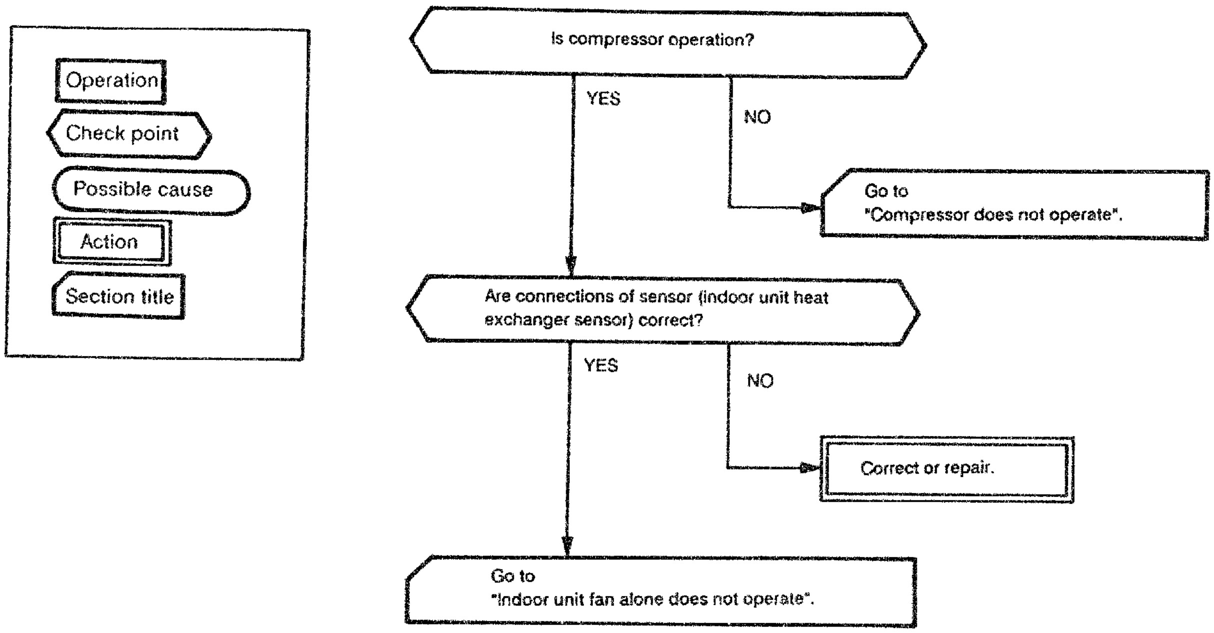
<Preliminary check>

(1) Is the outlet voltage 220/240V AC?
(2) Is the POWER SUPPLY switch on?
(3) Is the indoor unit fan dead in cool or fan only mode?

<Checking procedure>



3-5. Indoor unit fan is dead during heat operation mode.



3-6. Outdoor unit goes dead soon after start of operation.

<Preliminarly checks>

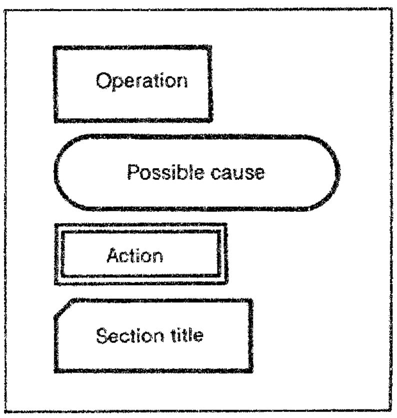
(1)Is the compressor made to stop by THERMO control?

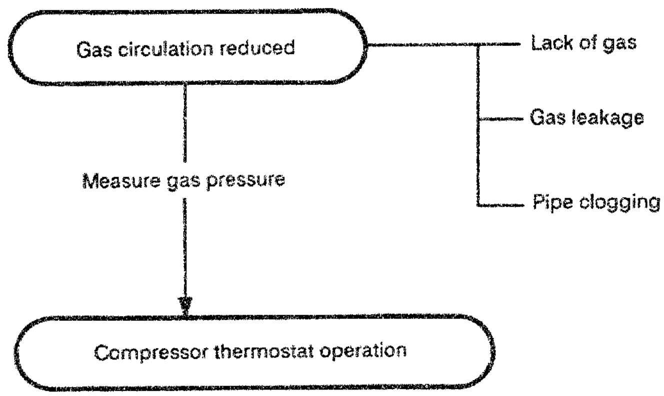
(2)Does 220/240V AC flow across ③ – ④ of the indoor unit microprocessor's terminal block and 11 – 50V DC across ④ – ⑤.

<Checking procedure>

Action varies depending on whether symptom <1> or <2> below occurs.

<1> OUTDOOR UNIT STOPS WITHIN 10 - 20 MINUTES AFTER START AND MORE THAN 10 MINUTES IS TAKEN TO RESTART.





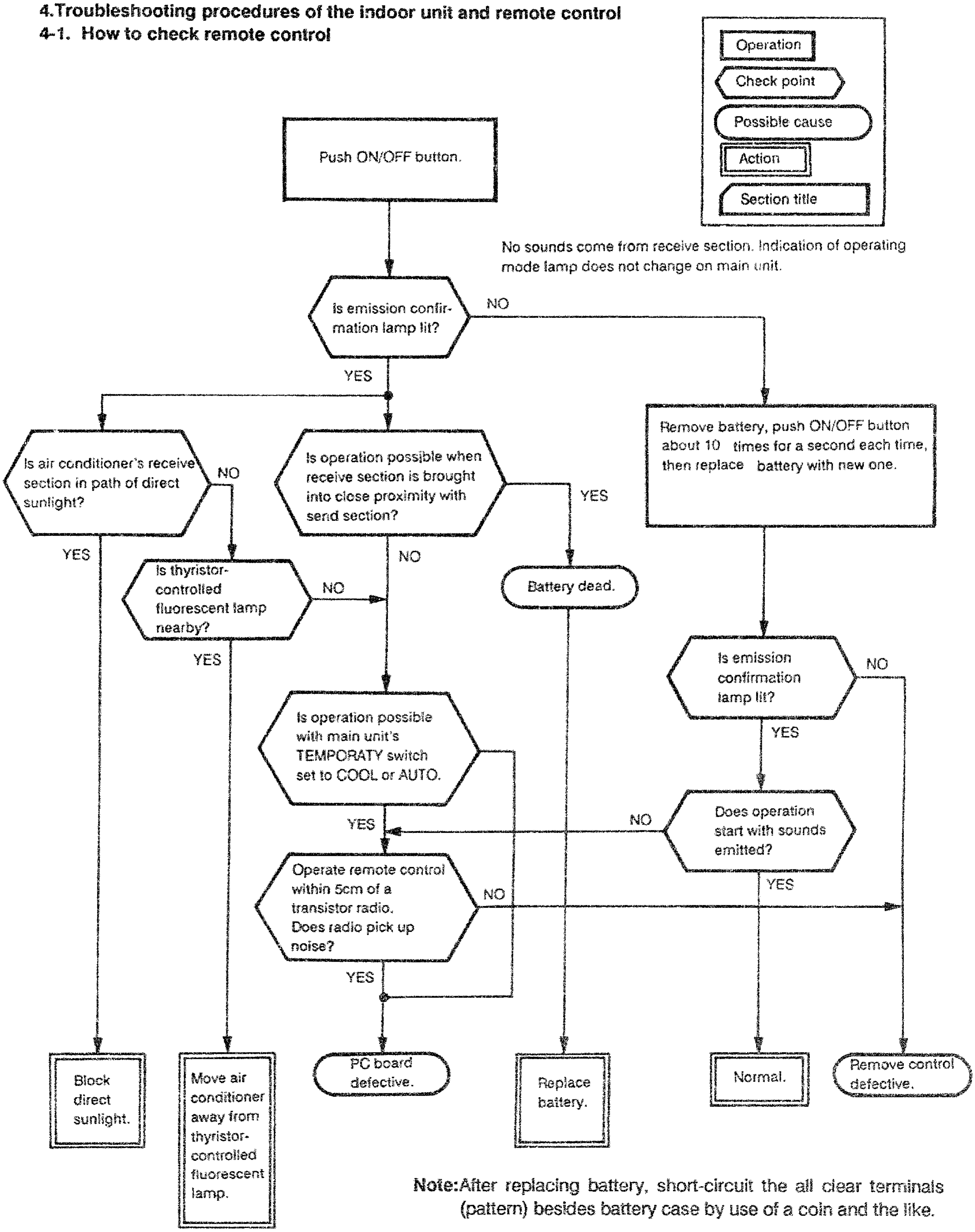
Check continuity across orange leads of outdoor unit fan's connector.

<2> ONCE OUTDOOR UNIT STOPS, IT DOES NOT RESTART WITHOUT POWER-ON SEQUENCE PERFORMED AGAIN.

Go to 'Compressor does not operate'.

4.Troubleshooting procedures of the indoor unit and remote control

4-1. How to check remote control



Note: After replacing battery, short-circuit the all clear terminals (pattern) besides battery case by use of a coin and the like.

4-2. How to check the printed circuit (PC) board

4-2-1. Precautions

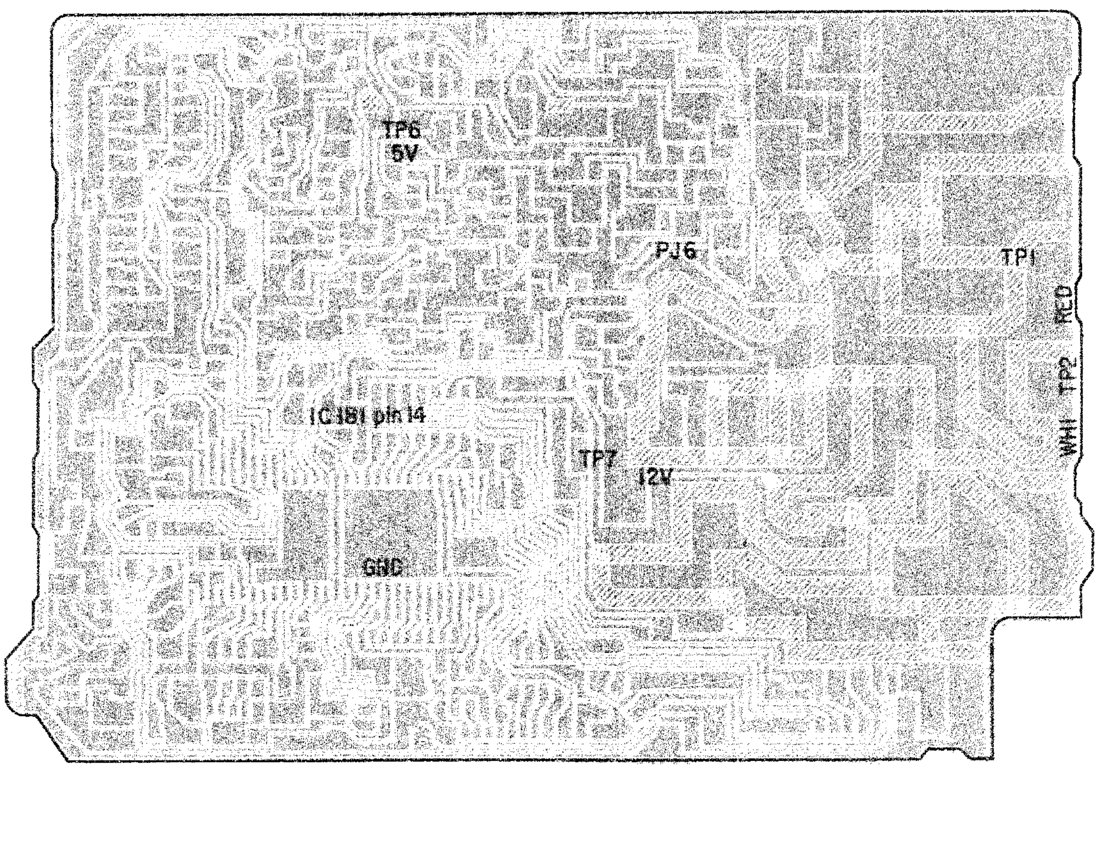
(1)Be sure to turn the power off before removing the front panel or the PC board.
(2)To remove the PC board, hold it at the edges so that no force is applied to the components.
(3)To unplug a connector from the PC board, do not pull the cable but hold the housing itself.

4-2-2. How to check

(1)If the PC board is suspected as being defective, check for disconnections of conductor wiring, burn-out and discoloration.
(2)The PC board consists of four sections as follows.

1) Main P.C. section – Power relay, motor power supply, motor drive circuit, control circuit.
2) Photosensor circuit section (Infrared rays receiving P.C. section) – Photosensor circuit.
3) Indicator section – LED.
4) TEMPORARY switch – TEMPORARY switch, AUTO LOUVER switch.

Follow the procedure given in Table 4 to check the PC board.



4-2-3. Checking procedure

Table 4

No.	Step	Checkpoint (symptom)	Couse of trouble
1	Turn off the main power. Remove PC board assembly from base. Disconnect cable from terminal block.	Fuse blown	(1) Impulsive voltage applied. (2) Motor power supply abnormal.
2	Unplug motor connector Turn on power. If ON lamp flashes (on and off in 0.5 second intervals), (1) – (3) at right need not be checked.	Check power-supply voltage (see Table 9). (1) TP1, TP2 (220/240V AC). (2) Across PJ6 (1), (3) (13V AC). (3) TP-6, GND (5V DC). (4) Q13 emitter, GND (12V DC).	(1) Power cord, POWER SUPPLY switch, fuse, and/or line filter defective or miswired. (2) Power transformer defective. (3) Power supply defective or load shorted. (4) Same as 3.
3	Push ON/OFF button once to enter operating mode, exclud-ing the FAN ONLY and TIMER ON modes.	Check power-supply voltage (see Table 9). (1) TP-7 (DC 12V), IC181 pin 14. (2) Across 3 – 4 of terminal block (220/240V AC).	(1) Disconnection in relay coil, relay driver (IC181) defective. (2) Relay contact failure, terminal block defective.
4	Start operation by shorting res-tart stop timer (see 4-2-4).	(1) All LEDs (ON, TIMER, M.F.C.) are lit. (2) Indications do not become normal after 3 seconds.	Indicator section or 4P housing as-sembly defective.
5	Push ON/OFF button once to enter operating mode. 1. Shortening restart stop timer setting (see Section 4-2-4). 2. COOL operation. 3. Air flow (SELECTOR) AUTO. 4. Preset temperature (THERMO) well below room temperature.	(1) Compressor does not run. (2) ON lamp is flashing (5Hz).	(1) Temperature unusually low in indoor unit heat exchanger. (2) Contact failure or disconnection of heat-exchanger sensor connector (off the position). (3) Heat-exchanger sensor or control PC board defective (See Table 11 to check resistance.) (4) Control PC board or indicator section defective.
6	Maintain condition of Step 5 and select the following conditions: 1. Heat operation. 2. Preset temperature well above room temperature.	(1) Compressor does not run. (2) ON lamp is flashing (5Hz).	(1) Temperature unusually high in heat exchanger. (2) Short of heat-exchanger sensor connector. (3) Heat-exchanger sensor defective (See Table 11 to check resistance.) (4) Control PC board or indicator section defective.
7	Connect motor connector to "motor" and turn on power. Select the following conditions and start operation: 1. Air flow (SELECTOR) FAN ONLY. 2. Fan "High". 3. Continuous operation (TIMER: CONTINUE).	(1) There develops a voltage of 120V or higher between Red and Black of motor. (2) Motor remains stationary (remote control keys are enabled). (3) Motor runs with considerable vibration.	(1) Indoor unit fan motor defective. (2) Contact failure in motor connector. (3) Main PC board defective.

Table 5 Supply voltages

No.	Multimeter lead: red (+)		Multimeter lead: black (–)		Requirement
1	TP-1 (P4)	Main PC board	TP-2 (P5)	Main PC board	220/240V AC (at power-on time)
2	PJ6 pin 1	↑	PJ6 pin 3	↑	13V AC (at power-on time))
3	Q12 (E) (TP-6)	↑	GND	↑	5V DC applied (at power-on time)
4	Q13 (E) (TP-7)	↑	GND	↑	12V DC applied (at power-on time)

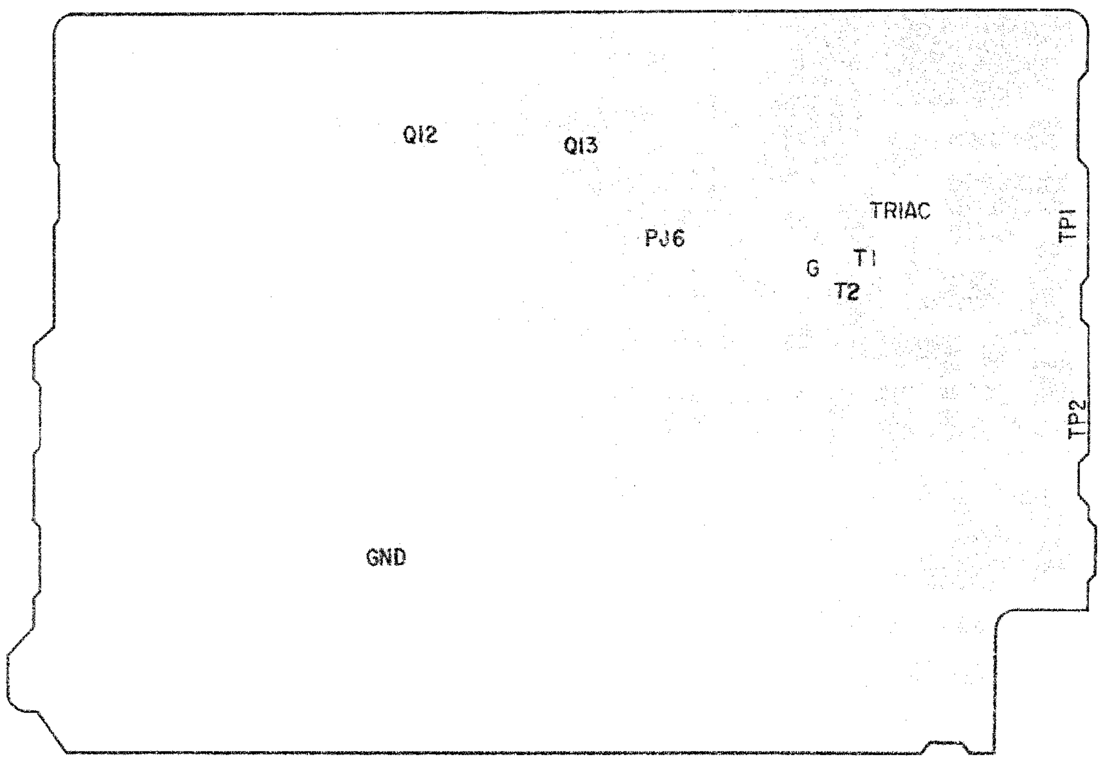


Table 6 Checkpoints of the motor power supply and drive TRIAC circuits
(The multimeter's battery must be 3V or more.)

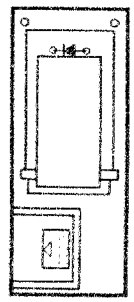
No.	Multimeter lead: red (+)		Multimeter lead: black (–)		Requirement
1	TR1 (T1)	Main PC board	TR1 (T2)	Main PC board	No continuity
2	TR1 (T2)	Main PC board	TR1 (G)	Main PC board	No continuity
3	TR1 (T1)	Main PC board	TR1 (G)	Main PC board	Continuity

Table 7 Resistance (approximate values) of sensor (thermistors)

(Unit: kΩ)

Sensor	0°C	10°C	20°C	30°C
Room temperature sensor (Thermo.)	35.8	20.7	12.6	8.0
Indoor heat-exchanger sensor	35.8	20.7	12.6	8.0
Outdoor heat-exchanger sensor	35.8	20.7	12.6	8.0

Wireless remote controller



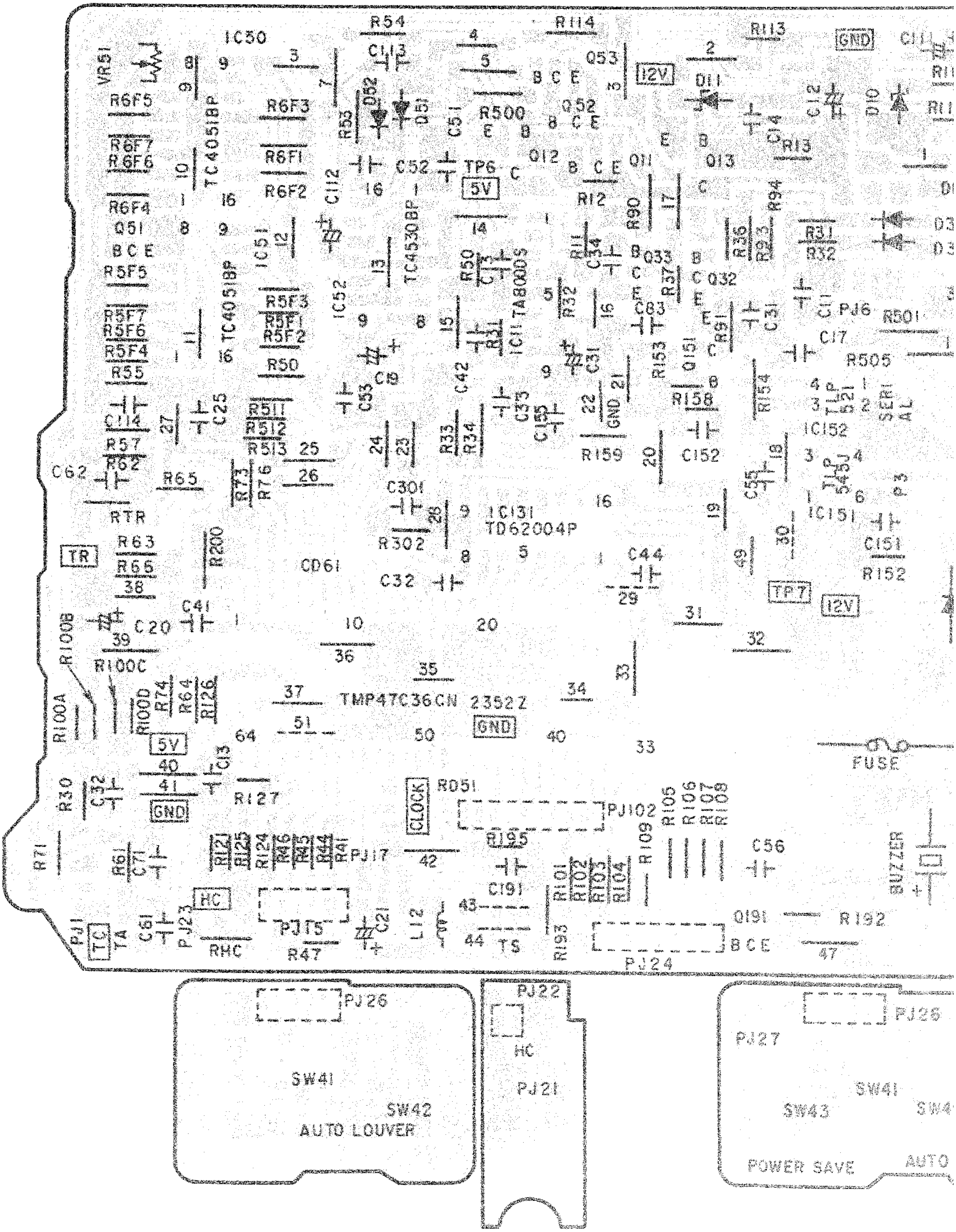
4-2-4. How to shorten the anti-restart timer setting

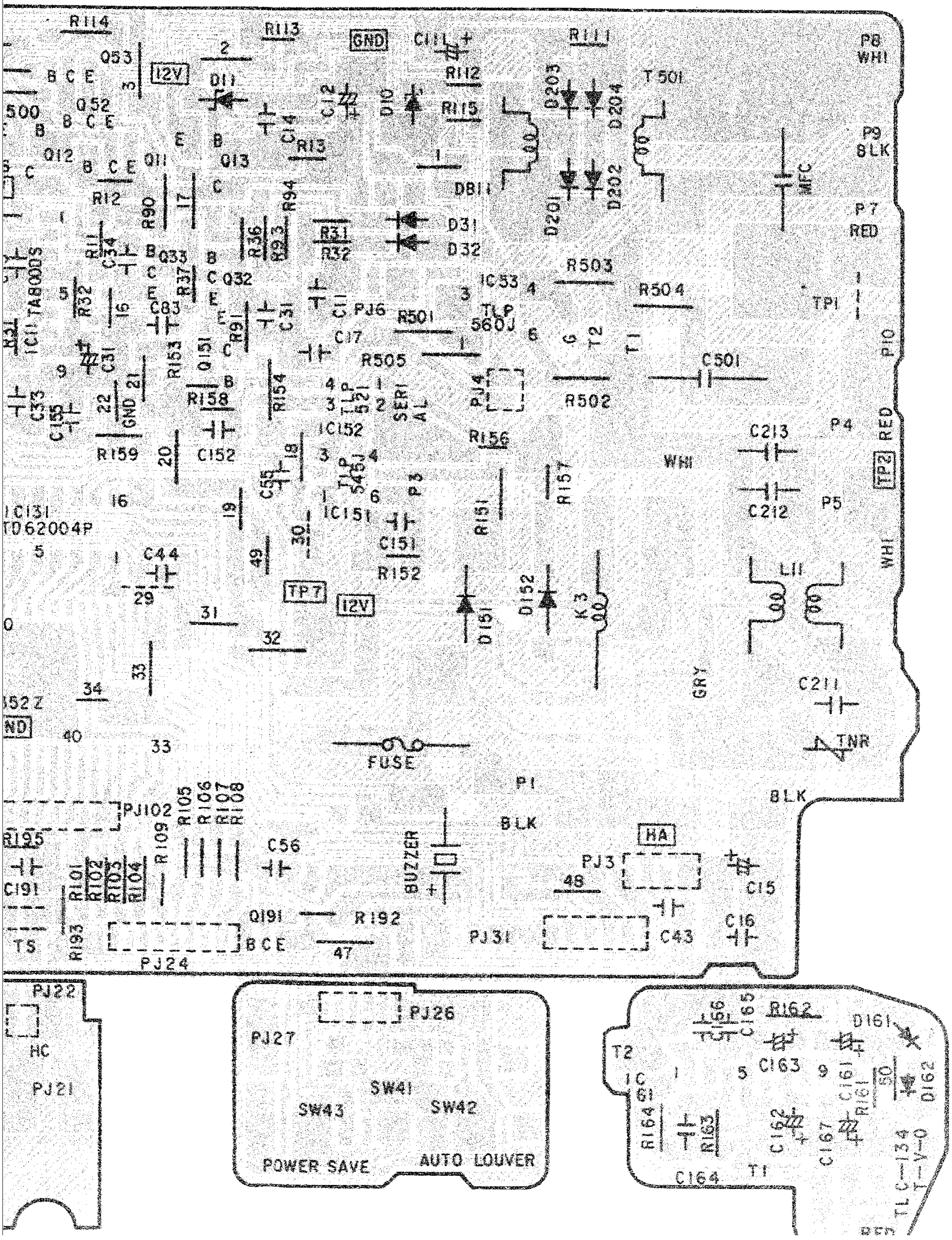
(1)There are two holes, with rivets inside, on the bottom of the remote control unit. Install a diode (IS1555 or equivalent) across the holes.

(2)At the beginning of operation, push the ON/OFF button with the diode installed. During the operation, push the RESET button with the diode installed. (Air flow is set to AUTO.)

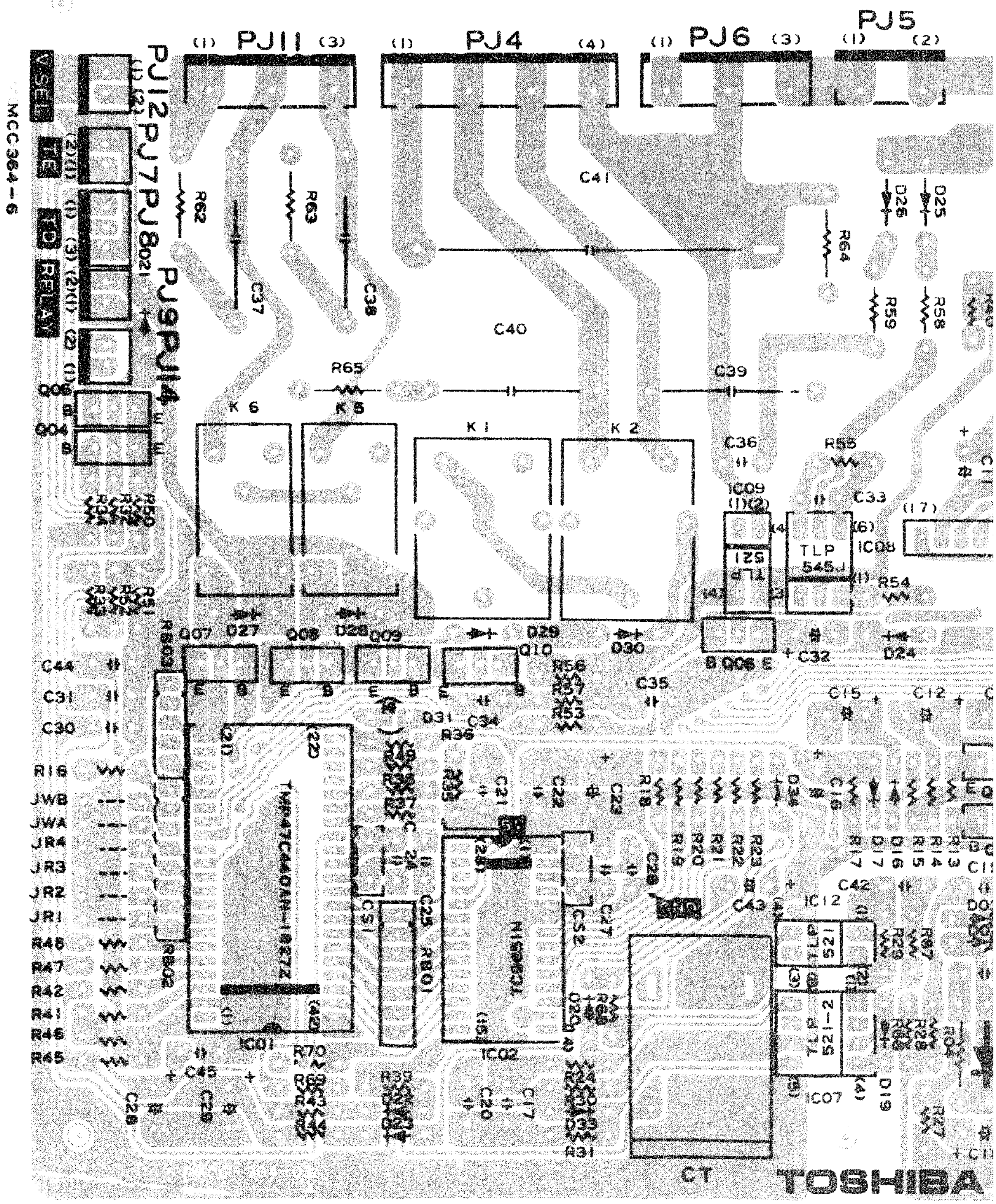
Note:

- If the outdoor unit is not connected.
- If you push the ON/OFF button when the compressor is on and the diode installed, operation stops in about 10 seconds as the power relay K8 turns off. (ON/OFF lamp flashes.) Then push the RESET button with the diode still installed and the timer setting is reduced. (The ON/OFF lamp continues to flash.)

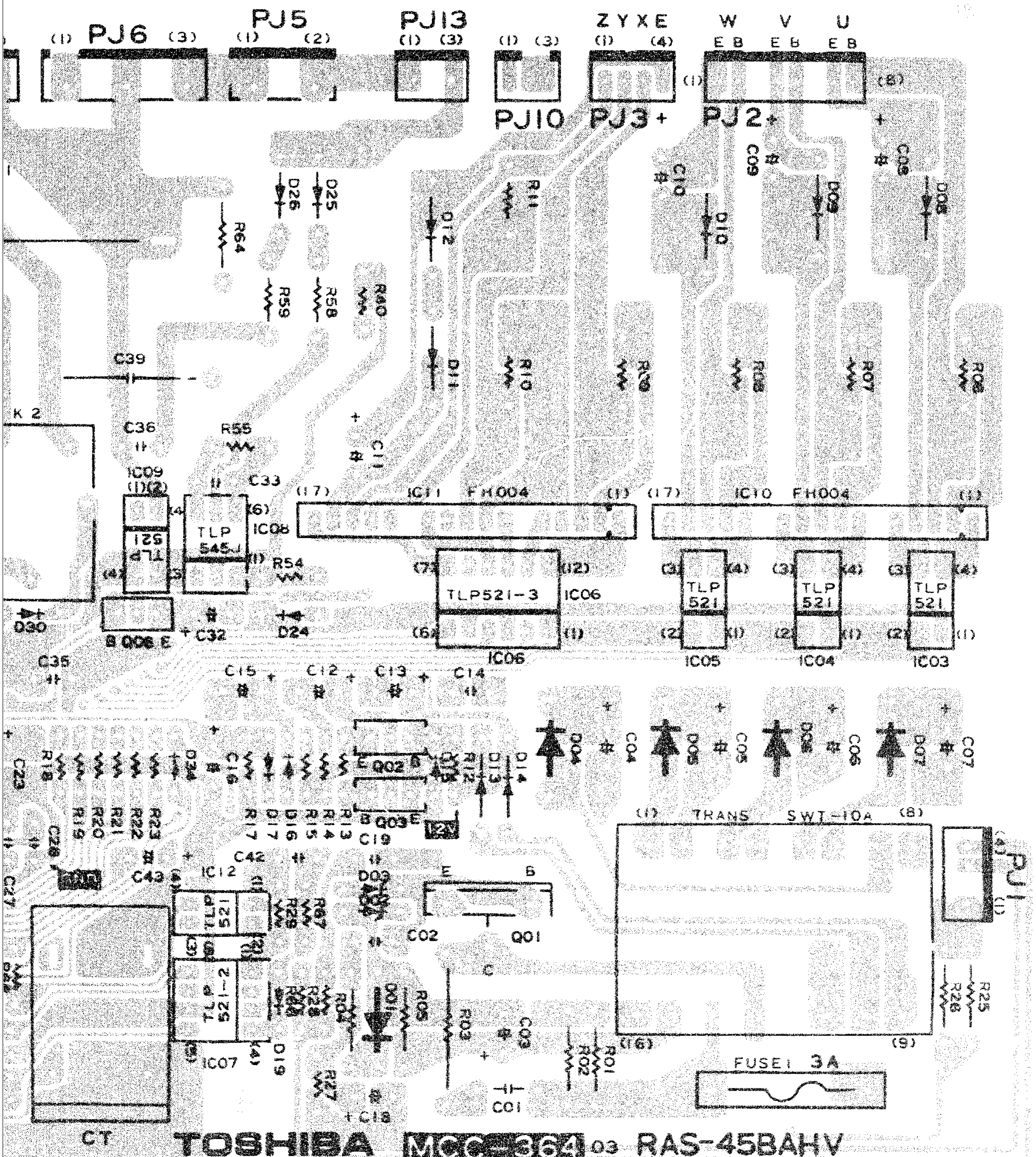

IC50
TC4051BP
R54
C113
D52
D51
R53
C51
R500
R114
Q53
12V
Q52
Q12
Q11
Q13
R113
GND
D11
C14
C12
D10
R13
R6F5
R6F7
R6F6
R6F3
R6F1
R6F2
R6F4
C112
C52
TP6
5V
R12
R90
R94
Q51
B C E
IC51
TC4530BP
TC4051BP
R5F5
R5F7
R5F6
R5F4
R55
R5F3
R5F1
R5F2
R50
C13
IC11 TA8000S
R32
R11
C34
Q33
R57
Q32
R36
R93
R31
R32
C83
R91
C31
C11
PJ6
R501
C17
R505
C19
C53
C42
R31
C31
R153
Q151
R158
R154
TLP521
SERIAL
C114
C25
R511
R512
R513
C33
C155
R159
C152
IC152
R57
R62
C62
R65
R73
R76
R33
R34
C55
TLP545J
IC151
P3
C301
IC131
TD62004P
R302
RTR
R63
R66
TR
R200
CD61
C32
C44
C151
R152
TP7
12V
R100B
C41
C20
R100
R100C
TMP47C36CN 2352Z
R100A
R100D
R74
R64
R126
5V
GND
C13
R30
C32
R127
CLOCK
R051
PJ102
R105
R106
R107
R108
R109
C56
BUZZER
R71
R61
C71
R121
R125
R124
R46
R45
R44
R41
PJ17
R195
C191
R101
R102
R103
R104
HC
PJ15
C21
L12
PJ1
TC
TA
C61
PJ23
RHC
R47
TS
R193
PJ24
Q191
B C E
R192
FUSE
PJ26
SW41
SW42
AUTO LOUVER
PJ22
HC
PJ21
PJ26
PJ27
SW41
SW43
POWER SAVE
AUTO

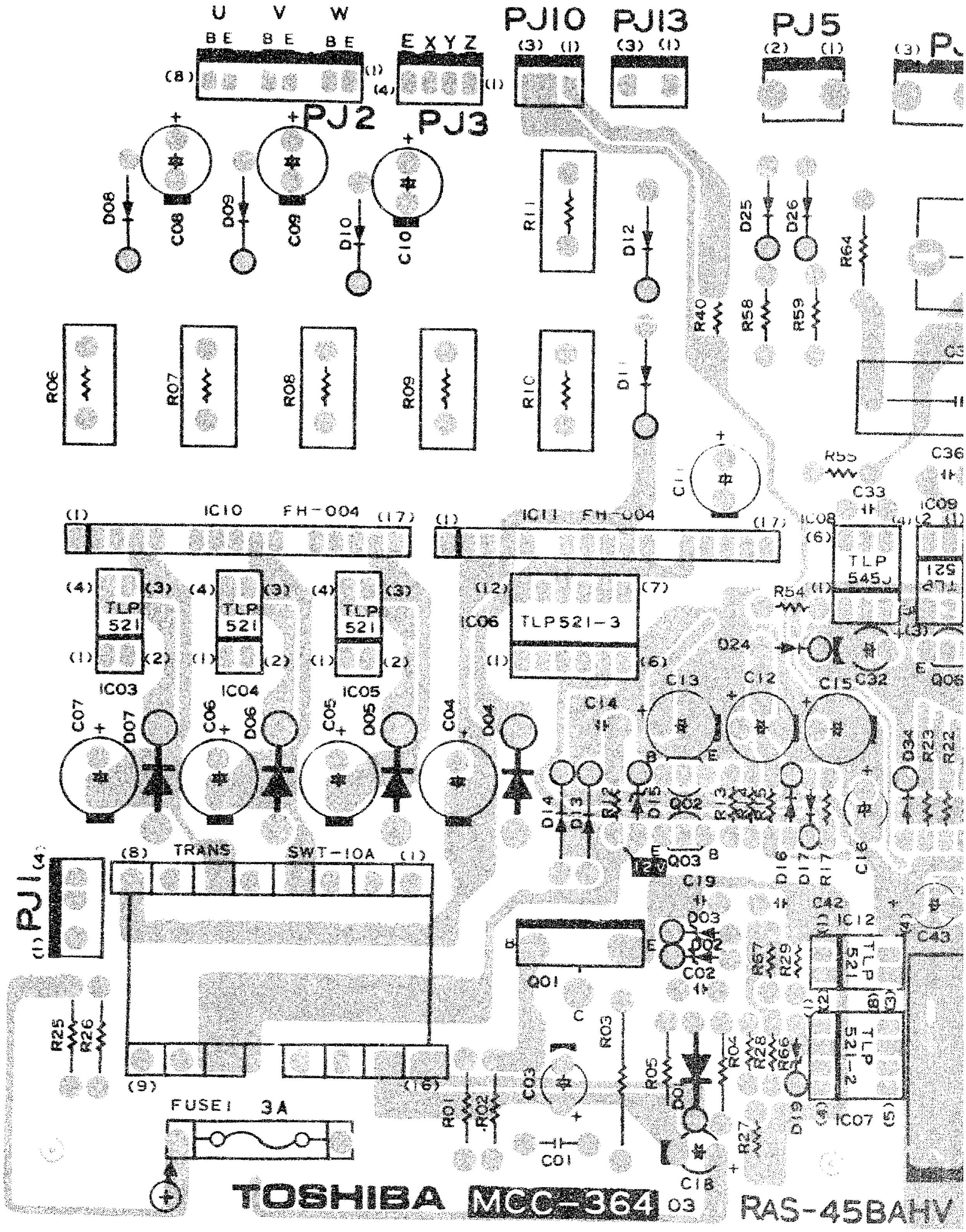


R114
R113
GND
C111
R111
P8 WHI
Q53
12V
D11
R112
T501
Q52
D10
R115
D203
D204
P9 BLK
Q12
Q11
Q13
R13
C14
C12
DB11
D202
D201
MFC
R12
R90
R94
R36
R93
R31
R32
D31
D32
P7 RED
Q33
Q32
R37
TA8005
C34
IC53
R503
R504
TP1
R32
C83
R91
C31
C11
PJ6
TLP 560J
R501
T2
T1
P10
IC11
C31
R153
Q151
C17
R505
C501
C33
C155
R158
R154
TLP 521
SERIAL
PJ4
R502
P4
RED
R159
GND
C152
IC152
R156
C213
WHI
TP2
TLP 545J
P3
R157
C212
P5
IC131
TD62004P
C55
IC151
R151
C44
C151
R152
L11
WHI
TP7
12V
D151
D152
K3
GRY
C211
TNR
FUSE
P1
PJ102
R105
R106
R107
R108
BLK
BLK
R195
R109
C56
HA
BUZZER
PJ3
C15
C191
R101
R102
R103
R104
Q191
R192
C16
C43
PJ31
R193
BCE
47
PJ24
PJ22
PJ26
R162
D161
PJ27
C166
C165
HC
C163
T2
IC61
PJ21
SW41
SW43
SW42
C161
R161
D162
R164
R163
C162
C167
TLC-134
T-V-0
POWER SAVE
AUTO LOUVER
C164
T1

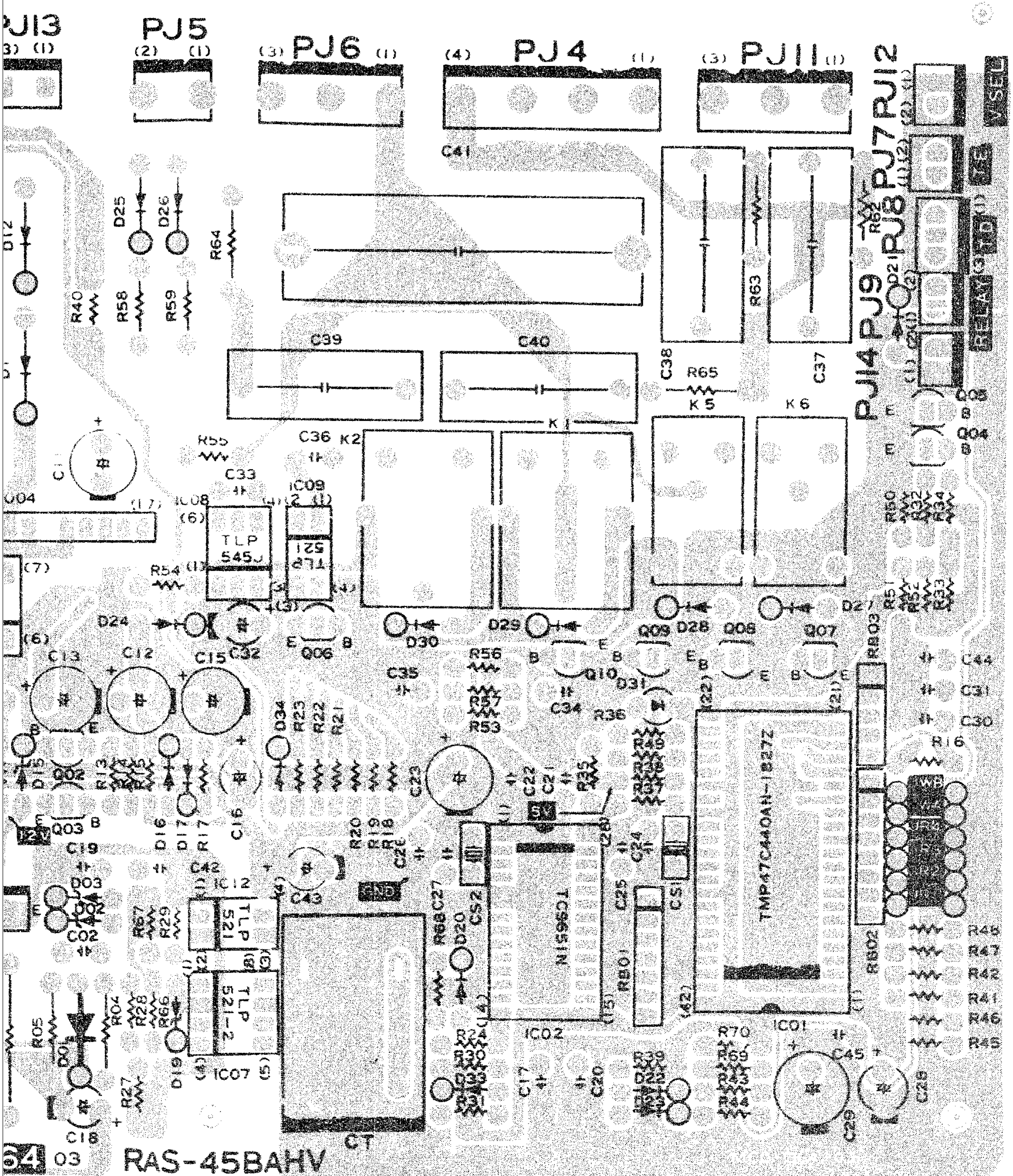


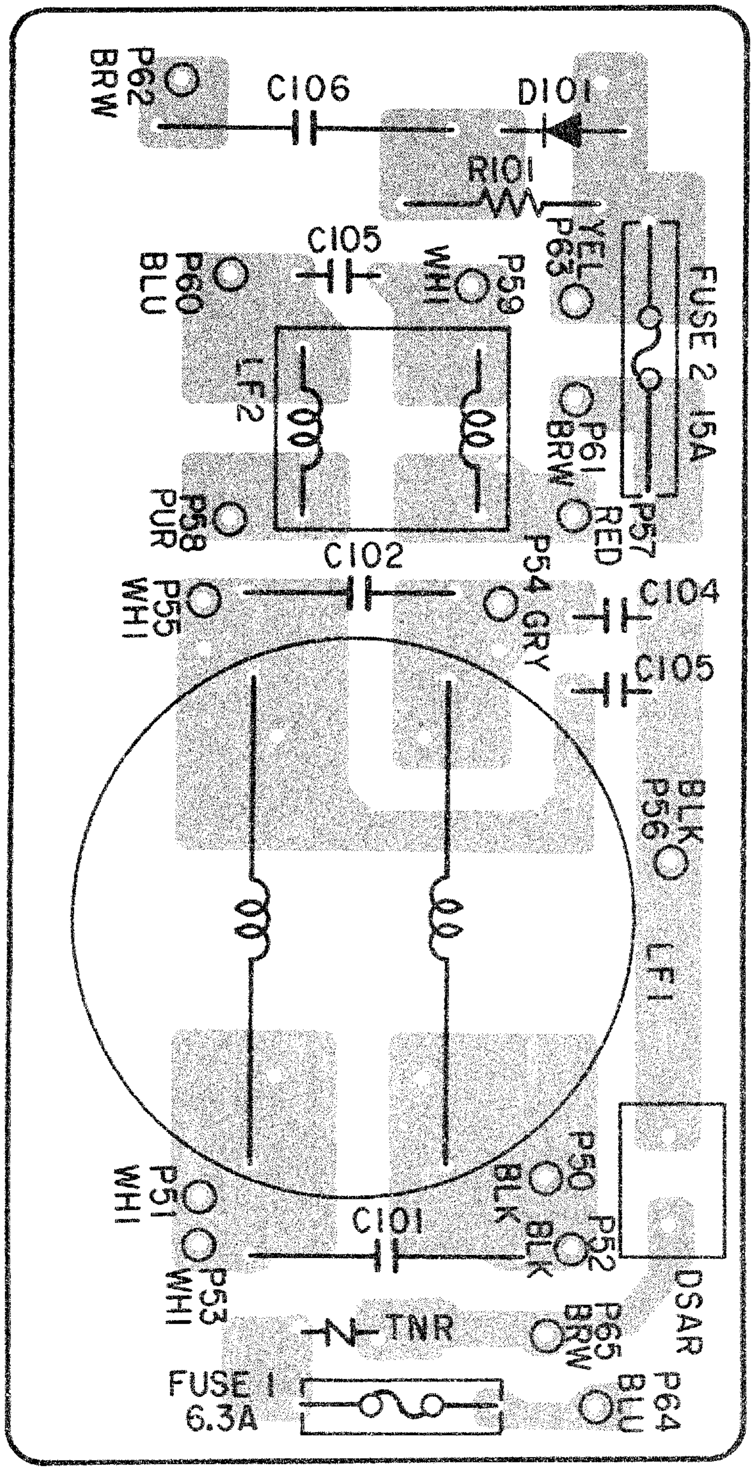
MCC 364-6
PJ11
PJ4
PJ6
PJ5
PJ12
PJ7
PJ8
PJ9
PJ14
V.SEL
T.E
F.D
RELAY
C41
C40
C39
C37
C38
R62
R63
R64
R65
R58
R59
D25
D26
K6
K5
K1
K2
C36
R55
IC09
C33
TLP 545J
IC08
R54
D27
D28
D29
D30
D24
Q04
Q05
Q06
Q07
Q08
Q09
Q10
C32
C35
C34
D31
R36
R56
R57
R53
C44
C31
C30
R16
JWB
JWA
JR4
JR3
JR2
JR1
R48
R47
R42
R41
R46
R45
RB03
RB02
RB01
TMP47C440AN-R872
IC01
IC02
TC9153N
CS1
CS2
C25
C21
C22
C23
C26
C27
C15
C12
C16
R18
R19
R20
R21
R22
R23
D34
R13
R14
R15
R17
D16
D17
IC12
C42
C43
R29
R97
TLP 521
TLP 521-2
IC07
D19
R27
C45
C28
C29
R70
R69
R43
R39
R31
C20
C17
Q20
CT
TOSHIBA


PJ6
PJ5
PJ13
PJ10
PJ3
PJ2
Z Y X E
W V U
E B E B E B
IC11 FH004
IC10 FH004
TLP 521
TLP 545J
IC08
IC09
TLP521-3
IC06
IC05
IC04
IC03
Q02
Q03
Q01
Q06
IC12
IC07
TLP 521-2
TRANS SWT-10A
PJ1
FUSE1 3A
CT
12V
TOSHIBA MCC-364 03 RAS-45BAHV



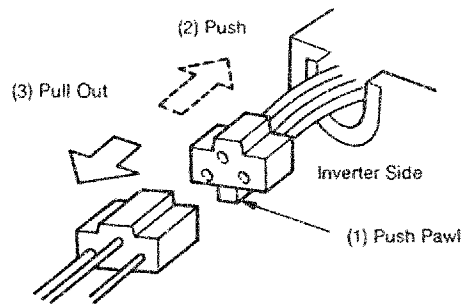
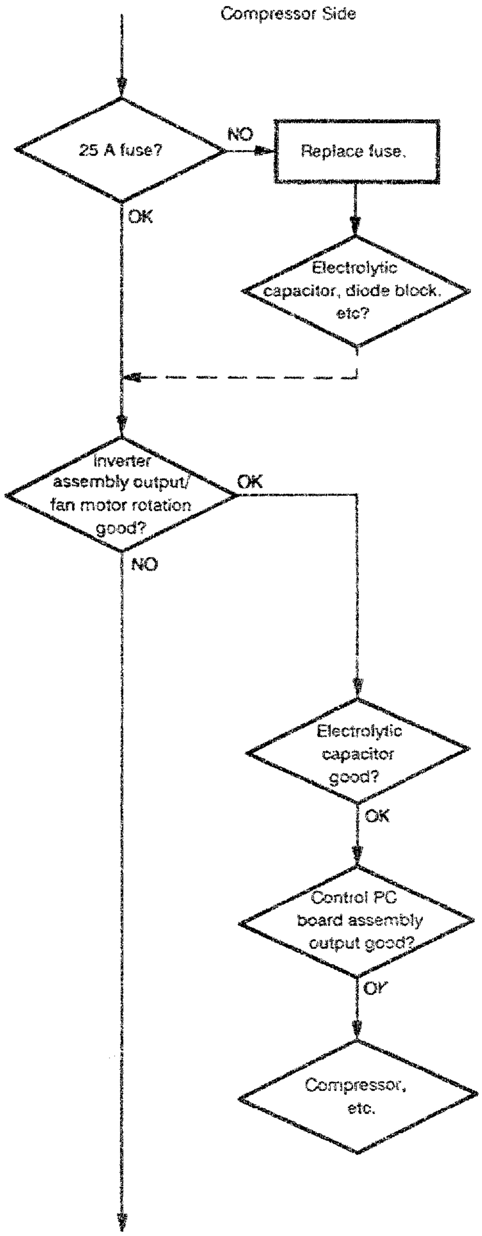
U V W
PJ10
PJ13
PJ5
PJ2
PJ3
E X Y Z
IC10 FH-004
IC11 FH-004
TLP 521
TLP521-3
TLP 545J
TRANS SWT-10A
PJ1
FUSE1 3A
TOSHIBA MCC-364 03
RAS-45BAHV

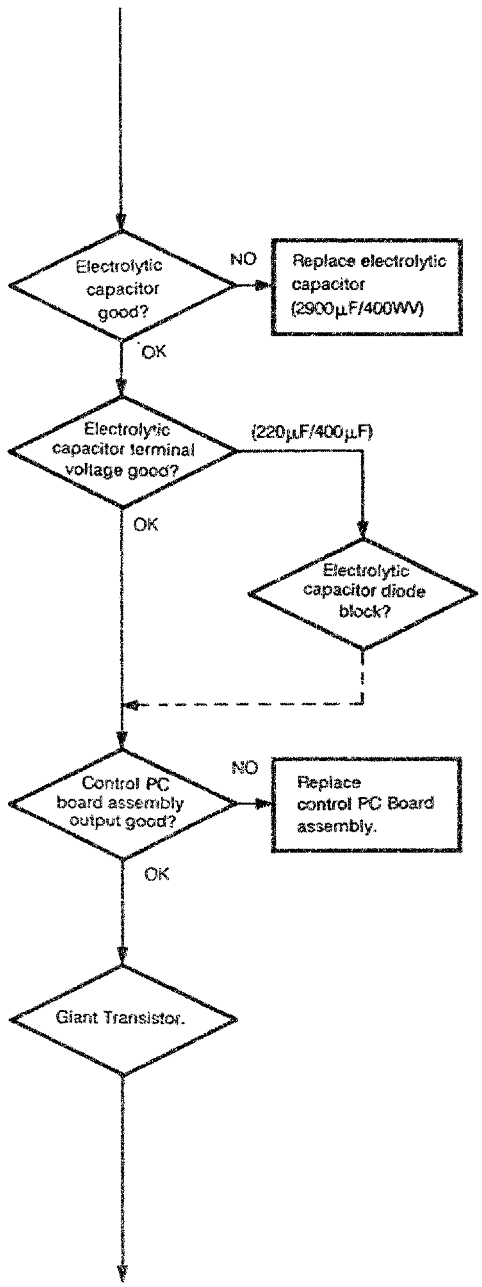
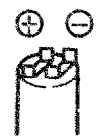
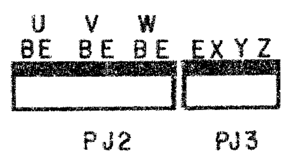

PJ13
PJ5
PJ6
PJ4
PJ11
PJ12
PJ7
PJ8
PJ9
PJ14
V SEL
RELAY
C39
C40
C37
C38
K1
K2
K5
K6
TLP 545J
TMP47C440AN-1827Z
TC9655N
IC01
IC02
IC07
IC12
CT
RAS-45BAHV



P62 BRW
C106
D101
R101
P63 YEL
FUSE 2 15A
C105
P60 BLU
P59 WHI
LF2
P61 BRW
P57 RED
P58 PUR
C102
P55 WHI
P54 GRY
C104
C105
P56 BLK
LF1
DSAR
P50 BLK
P51 WHI
C101
P52 BLK
P53 WHI
TNR
P65 BRW
FUSE 1 6.3A
P64 BLU

Table 8

No.	Flowchart	Remarks	Work category	Description
1		Preparation	(1) Cut off power. (2) Using a soldering iron's input connector, discharge electirciyty across (1), (2), (3) and (4) of terminal block, and across compressor lead terminals. (3) Disconnect connector between inverter and compressor (keeping pawl pushed in, push connector once then pull out).	**CAUTION** To let capacitor discharge into noise filter circuit.
		Check	Check that 25A fuse located beside terminal block is not blown.	If fuse is blown, be sure to check diode block and electrolytic capacitor.
		Opearation	Shorten the anti-restart time then operate outdoor unit fan in cool or heat operation mode.	To output voltage across compressor leads.
		Measurement	Measure voltage across compressor leads at inverter-side connector.	Voltage must be steady (+/−2V or so) at levels shown in Table 12 and outdoor unit fan motor must run properly.
2		Preparation	(1) Cut off power and let capacitor discharge. (2) Remove yellow and brown leads from transistor module assembly. (3) Disconnect connectors 4P (PJ3) and 8P (PJ2) of giant transistor from control PC board assembly. (4) Remove blue and brown lead from electrolytic capacitor (2900μF/400WV).	**CAUTION** Check voltage using the multimeter after the capacitor is discharged. (10V or less)

No.	Flowchart	Remarks	Work category	Description
3		Operation	Shorten the anti-restart timer then operate compressor in cool operation mode.	
4		Measurement	(1) Measure terminal voltages of two electrolytic capacitors (220µF/400WV).	Must be 330-380 VDC.
5				
6			(2) Measure DC voltage across output terminals of control PC board assembly.	Good if reading is 2-3V DC. Control PC board assembly must fail due to failure of giant transistor. So, always check giant transistor when control PC board.
			(3) Measure voltage at connector with plug removed from control PC board assembly.  Measure at 6 points with E connected to (–).	
		Stop	Turn off power, then discharge two capacitors using a soldering irons.	
		Check	Check giant transistor.	
		Complete end	Restore initial condition.	

5. Troubleshooting the inverter assembly

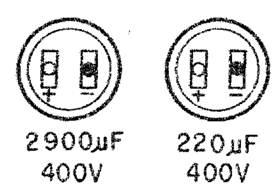
Table 9 Operation frequency and inverter output voltage

Frequency	Inverter output voltage (AC)under no load (V)
120Hz	235 – 255
100Hz	230 – 250
75Hz	180
50Hz	140
35Hz	110
OFF	0

Notes:

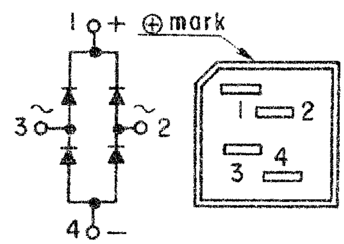
1. The power-supply voltage is assumed to be 220/240V.
2. The "no load" condition means that the leads connecting to the compressor are removed.
3. Note, that when the inverter output leads are connected to the compressor, voltage readings are lower than as specified above.
4. The 300V AC range of a multimeter is assumed for the above inverter output voltages.

Table 10 How to check parts

<table>
<tr><th>No.</th><th>Part</th><th>Checking procedure</th></tr>
<tr><td>1</td><td>Capacitor</td><td>
1. Cut off power.
2. Discharge all two electrolytic capacitors completely.
(Make sure capacitor voltage across ⊕ –⊖is less than 10V.)
3. Remove all minus-side leads.
4. Make sure safety valve is intact.
5. Make sure container is not swollen and has not burst.
6. Make sure electrolyte is not leaking.
7. Make sure capacitor appears to be charged properly by performing the multimeter continuity test.


<table><tr><th>Condition with good capacitor</th></tr><tr><td>Meter pointer swings then returns slowly. Reverse meter lead polarity and make sure meter pointer moves as before.</td></tr></table>
</td></tr>
</table>

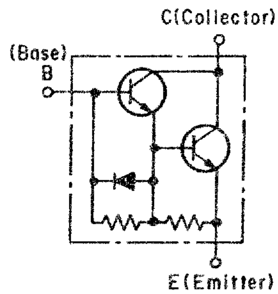
No.	Part	Checking procedure
2	Diode block	1. Cut off power. 2. Discharge all four electrolytic capacitors completely. (Make sure capacitor voltage across ⊕ – ⊖is less than 10V.) 3. Remove all four fastener connectors. 4. Make sure capacitor appears to be charged properly by performing the multimeter continuity test.

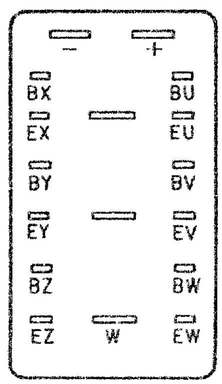


When meter lead polarity is reversed 10 – 20 ohms

Multimeter leads		Good resistance
⊕	⊖	
+	~	∞
	~	
−	~	
	~	

No.	Part	Checking procedure
3	Transistor module (continued)	1. Cut off power. 2. Discharge all four electrolytic capacitors completely. (Make sure capacitor voltage across ⊕ −⊖is less than 10V.) 3. Remove transistor module from inverter. 4. Check C-E current leak, V_{BE} and other characteristics of six transistors using the multimeter.



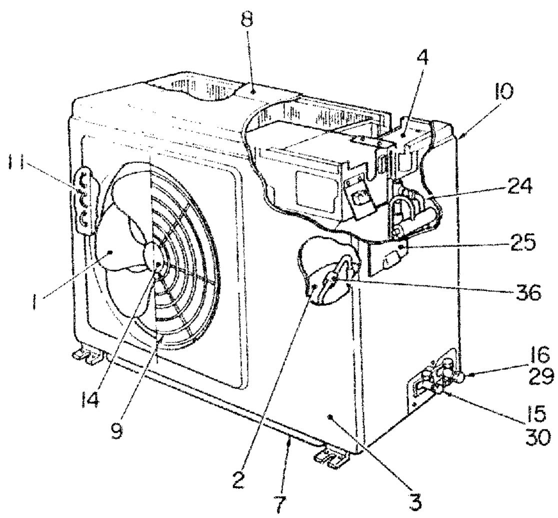


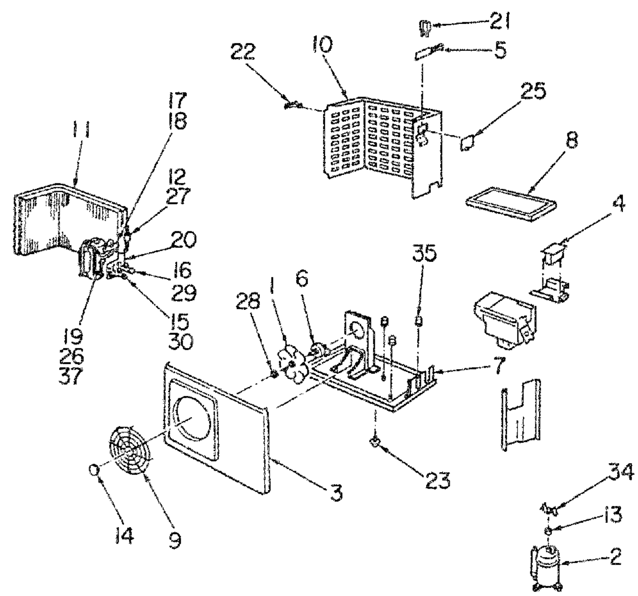
Multimeter leads		Good resistance
⊖	⊕	
+	−	∞
	BU	∞
	EU	∞
	BV	∞
	EV	∞
	BW	∞
	EW	∞
	U	∞
	V	∞
	W	∞
U	BX − EX	∞
V	BY − EY	∞
W	BZ − EZ	∞

When meter lead polarity is reversed Approx. 10 to 300 ohms

11. EXPLODED VIEWS AND PARTS LIST

1. Outdoor unit

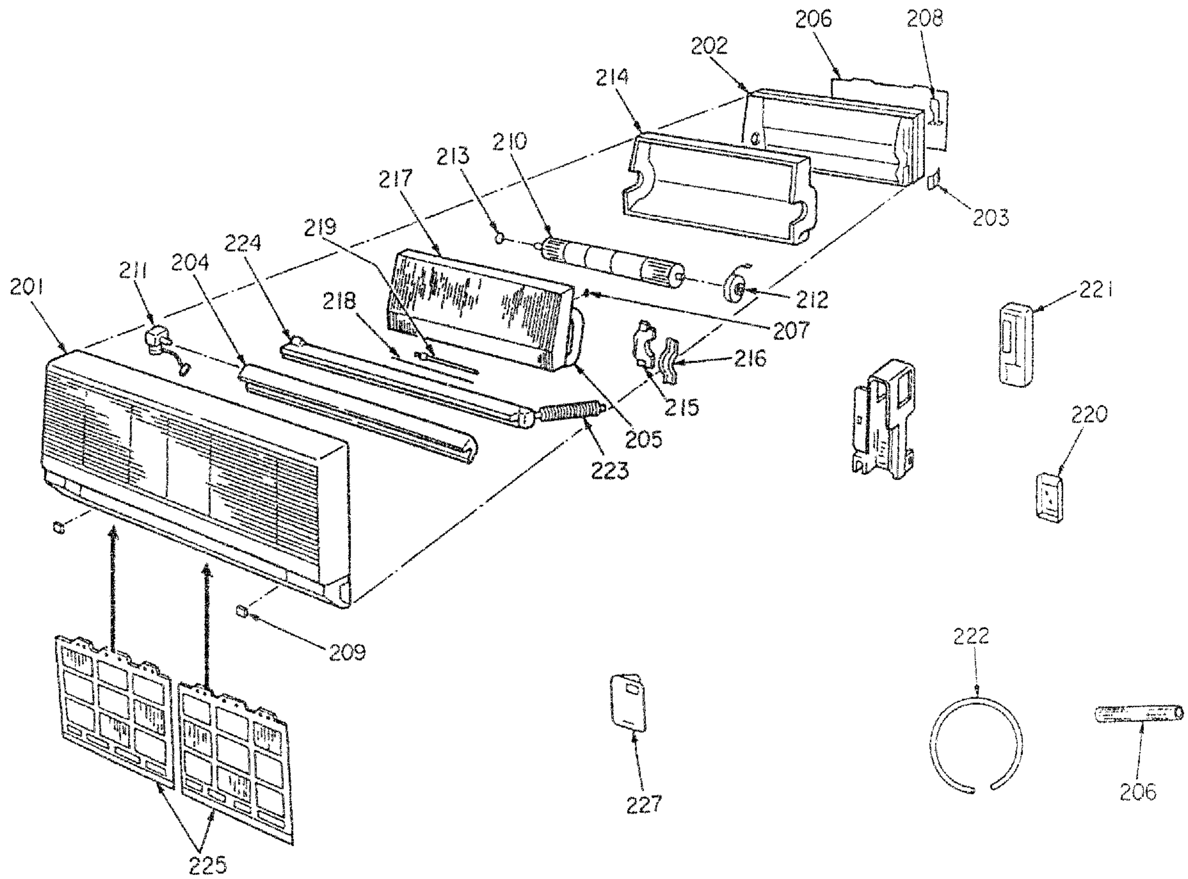




Location No.	Part No.	Description
1	43020208	FAN, PROPELL,360
2	43041021	COMPRESSOR, HV340 X 2-S12L1
3	43000486	CABINET, FRONT
4	43058176	REACTOR, CH-17
5	43050332	SWITCH, BIMETAL
6	43021829	'MOTOR, AC, 50Hz, 50W, FAN
7	43042406	BASE
8	43000569	CABI-UP
9	43007845	GUARD, FAN
10	43000578	CABINET-SIDE
11	43043514	CONDENSOR
12	43046182	2-WAY VALVE
13	43050298	THERMOSTAT, BIMETAL
14	43001495	MARK KOKAGE
15	43046228	PACKED VALVE 6.35
16	43046235	PACKED-VALVE, 12.7
17	43046216	SOLENOID, COIL
18	43146368	4-WAY VALVE
19	43046222	EXPANSION VALVE

Location No.	Part No.	Description
20	43045028	DRYER
21	43063145	HOLDER, THERMO
22	43019736	HANDLE
23	43032198	NIPPLE, DRAIN
24	43063210	HOLDER, SENSOR
25	43002108	COVER, ELECTRIC, PARTS
26	43049652	TUBE, SPIRAL
27	43046198	COIL, 2-WAY VALVE
28	43022347	FLANGE, NUT
29	43147195	BONNET 12.7
30	43147196	BONNET 6.35
31	43047491	CAPILLARY TUBE ø1.5
32	43047527	CAPILLARY TUBE ø2.0
33	43047584	CAPILLARY TUBE ø0.6
34	43063195	HOLDER THERMOSTAT, BIMETAL
35	43049625	CUSHION, RUBBER
36	43063198	HOLDER, SENSOR
37	43049651	HOLDER

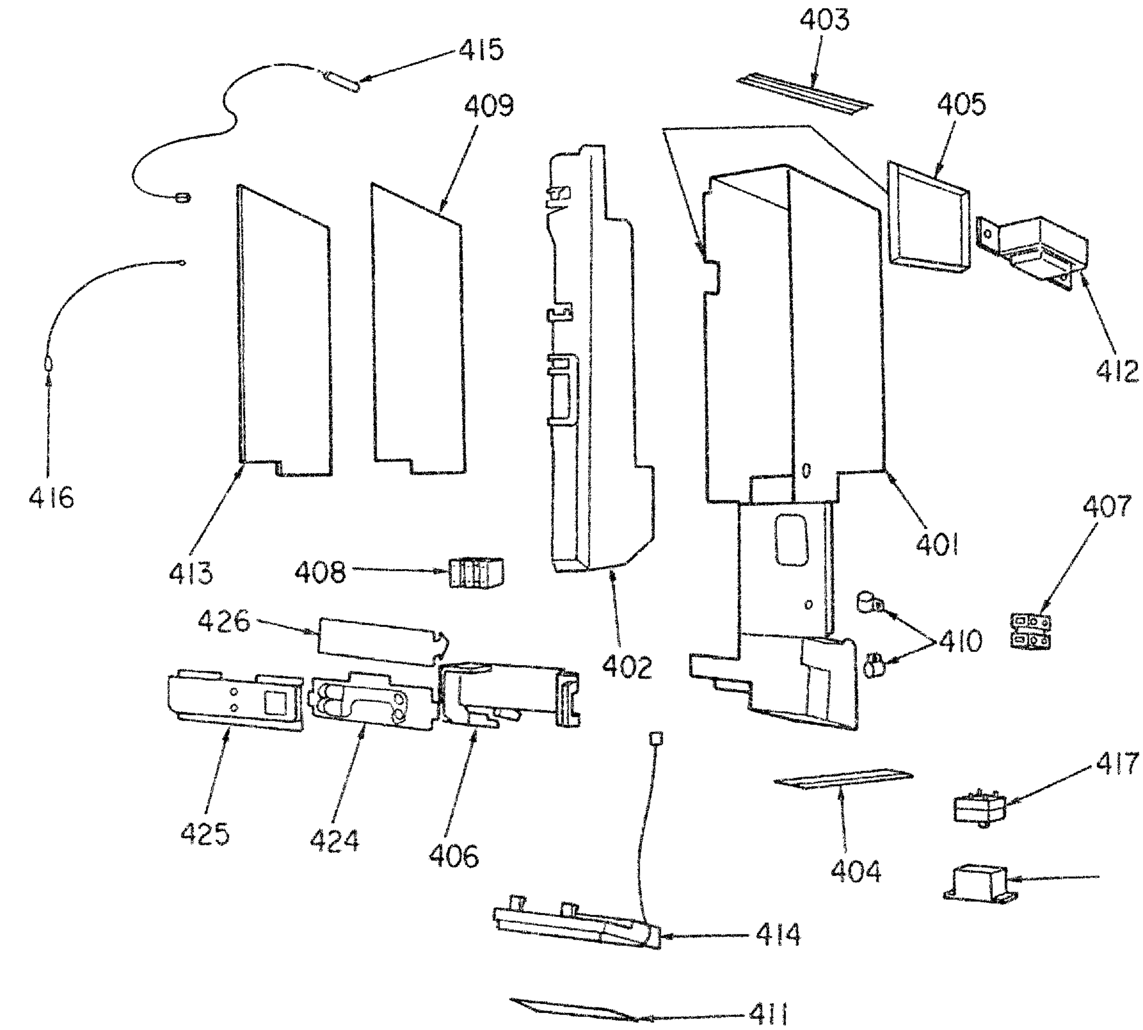
2. **Indoor unit**



Location No.	Part No.	Description
201	43000571	FRONT PANEL
202	43003171	REAR PLATE
203	43007665	BUSH, REAR PLATE
204	43007666	GRILLE
205	43011498	PIPE, SEALED
206	43011499	PIPE, SEALED
207	43019604	HOLDER, SENSOR
208	43019668	HOLDER, PIPE
209	43019706	CAP, SCREW
210	43020216	FAN, CROSS FLOW
211	43021699	MOTOR LOUVER
212	43021826	FAN MOTOR
213	43022321	BEARING
214	43022368	FAN CASE

Location No.	Part No.	Description
215	43039227	MOTOR BAND, LEFT
216	43039228	MOTOR BAND, RIGHT
217	43044522	EVAPORATOR
218	43047331	PIPE, DELIVERY
219	43047334	PIPE, SUCTION
220	43063209	HOLDER, REMOTE CONTROLLER
221	43069580	REMOTE CONTROLLER
222	43070090	HOSE, 3M, DRAIN
223	43070140	HOSE, DRAIN
224	43072304	DRAIN PAN AND GRILLE
225	43080275	AIR FILTER
226	43082245	PLATE, INSTALLATION
227	43085519	OWNER'S MANUAL AND REMOTE CONTROLLER MANUAL

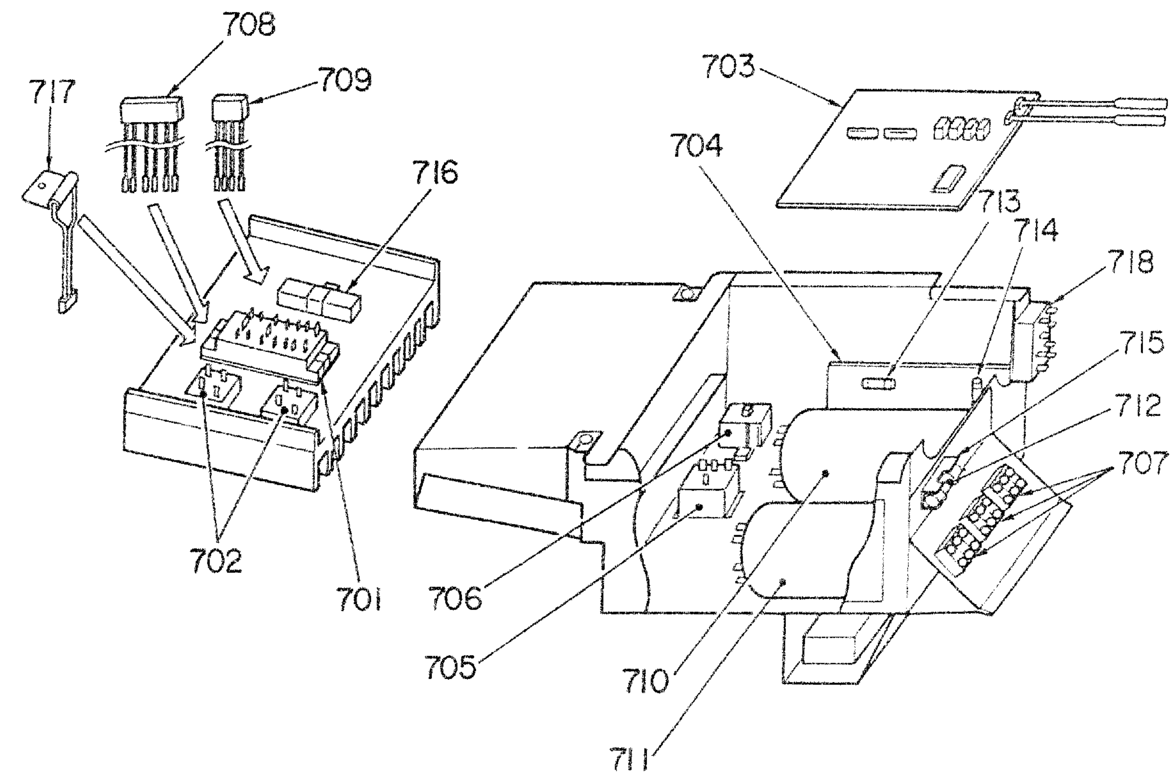
3. Control unit



Location No.	Part No.	Description
401	43061160	BASE, E-PARTS ASS'Y
402	43061159	GUIDE, DRAIN
403	43063244	GUIDE, P.C. BOARD, UPPER SIDE
404	43063245	GUIDE, P.C. BOARD, LOWER SIDE
405	43063243	COVER, TRANSFORMER
406	43061156	HOLDER LED
407	43060778	TERMINAL-BLOCK
408	43060794	TERMINAL-BLOCK
409	43061154	SHEET, INSULATION
410	43063181	CRAMP, CORD
411	43008204	PANEL, SWITCH
412	43058220	TRANSFORMER, POWER, FT25
413	43069617	P.C. BOARD ASS'Y
414	43050327	SENSOR ASS'Y, THERMAL RADIATION

Location No.	Part No.	Description
415	43050325	SENSOR, HEAT EXCHANGER
416	43050330	SENSOR, THERMOSTAT
417	43051315	SWITCH, POWER
418	43069259	FEED-BACK-TRANS-AC230V
419	43055411	CAPACITOR, METALIZED POLYESTER F1 LM
420	43055359	TOSHIBA NONLINER RESISTOR
421	43055415	CAPACITOR, SUPRESSING, CK
422	43055416	CAPACITOR, PLASTIC FILM, FADD
423	43060884	FUSE, 3A, 250V
424	43069319	P.C. BOARD ASS'Y, MCC-278
425	43007601	BASE, LED
426	43069323	SHEET, INSULATOR
427	43061157	HOLDER, SWITCH

4. Inverter unit



Location No.	Part No.	Description
701	43031084	TRANSISTOR MODULE, MG30G6EL2
702	43169388	DIODE BLOCK
703	43069635	P.C. BOARD ASS'Y, MCC-364
704	43069642	P.C. BOARD ASS'Y, MCC-369
705	43054276	RELAY, DC
706	43053045	RESISTOR, POSITIVE
707	43060794	TERMINAL BLOCK,3P
708	43069347	CONNECTOR WITH LEAD(8P)
709	43069349	CONNECTOR WITH LEAD (4P)
710	43055382	CAPACITOR, ELECTOROLYTIC 2900MFD, 400V

Location No.	Part No.	Description
711	43055387	CAPACITOR ELECTROLYTIC 220MFD, 400V
712	43160296	FUSE, 25A, 250V
713	43060726	FUSE, 15A, 250V
714	43060761	FUSE, 6.3A, 250V
715	43160298	FUSE BLOCK
716	43055383	RESISTOR 45mΩ
717	43069648	TRS-THERMOSTAT
718	43060776	TERMINAL BLOCK, 2P

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